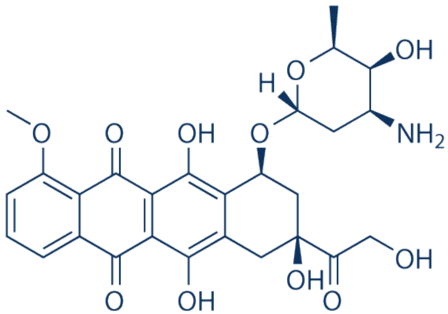
Drug delivery – Drugs

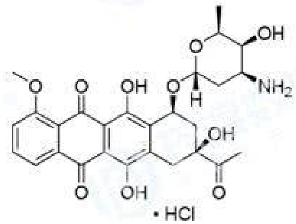


.HCl

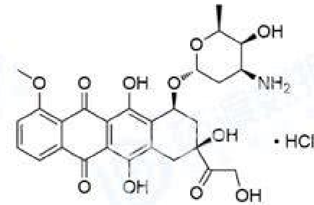
Doxorubicin (Adriamycin) HCl

亚德里亚霉素，阿霉素，

盐酸多柔比星

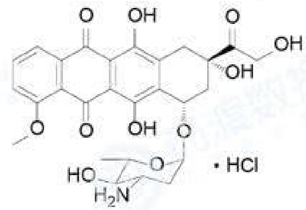


盐酸柔红霉素

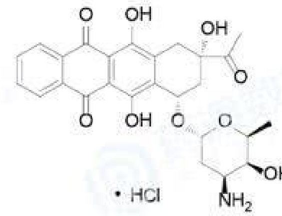


盐酸多柔比星

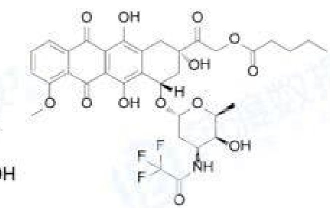
吡柔比星



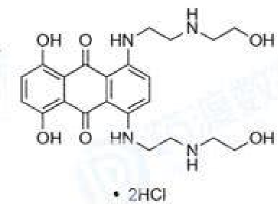
盐酸表柔比星



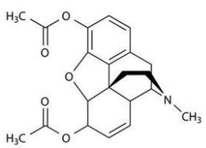
盐酸伊达比星



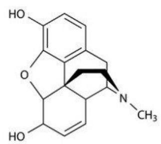
戊柔比星



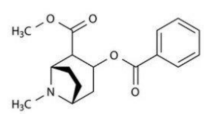
盐酸米托蒽醌



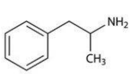
Heroin



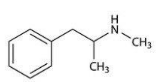
Morphine



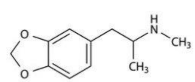
Cocaine



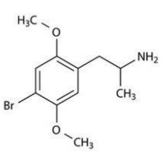
Amphetamine



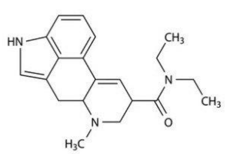
Methamphetamine



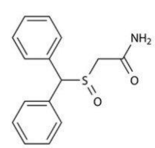
Ecstasy (MDMA)



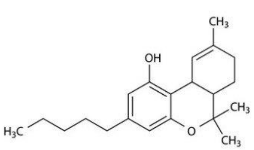
DOB



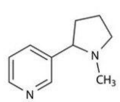
LSD



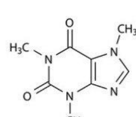
Modafinil



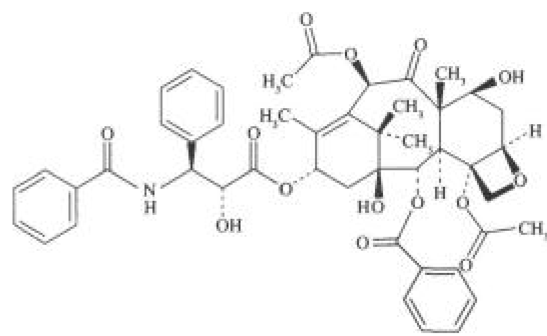
Tetrahydrocannabinol



Nicotine

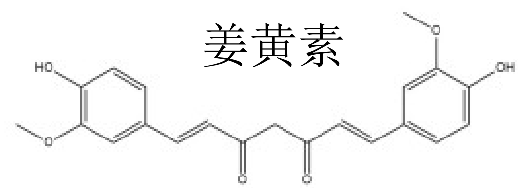


Caffeine



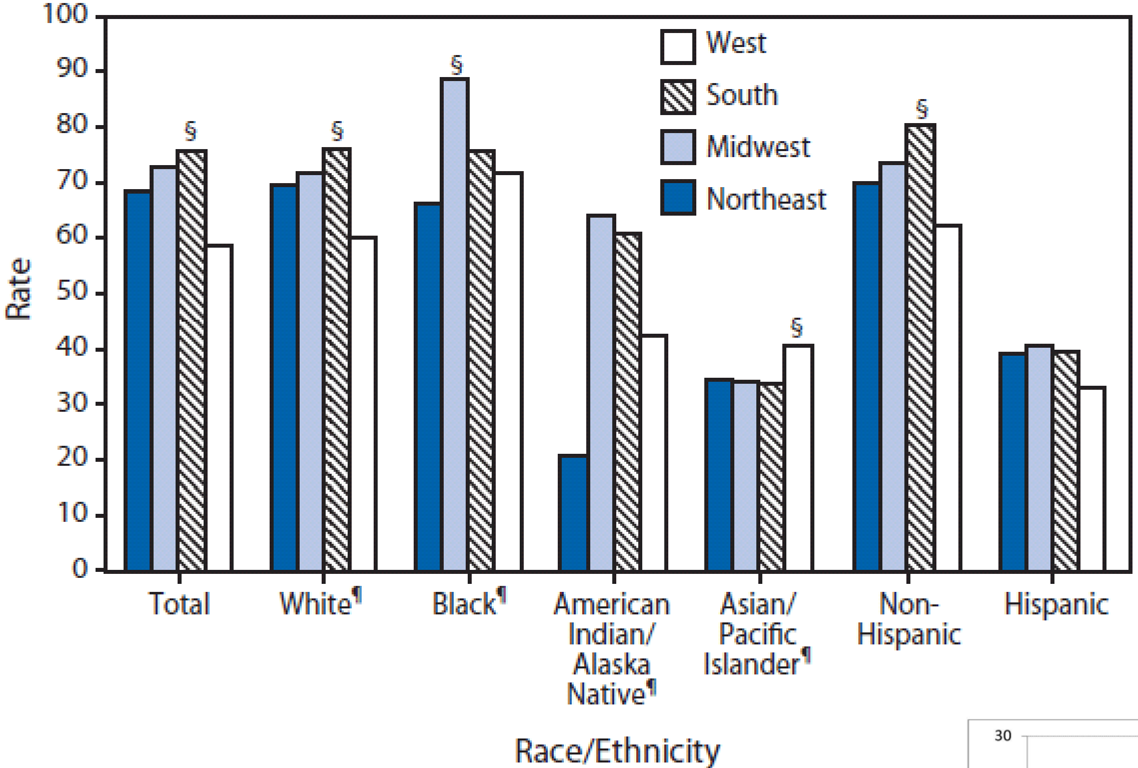
Paclitaxel 太平洋紫杉醇

姜黄素



"The siRNA or small interfering RNA is a 22 to 25 basepair long smaller molecules of RNA having a dinucleotide overhang at the 3', interfere in the protein synthesis by blocking the translation."

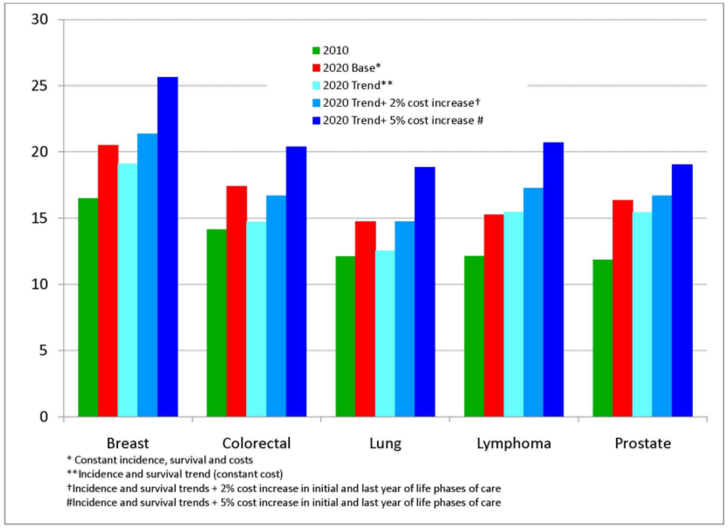
Drug delivery – Race/Ethnicity/Diseases



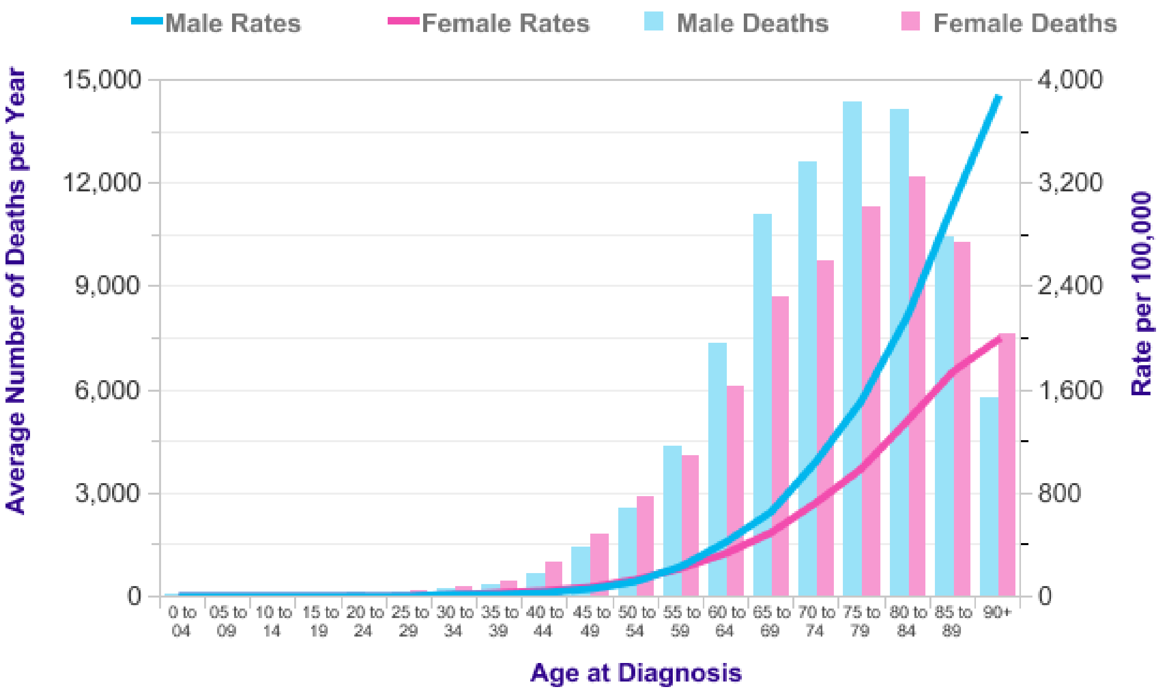
http://oncozine.com/profiles/blogs/medical-expenditure-for-cancer

Lung cancer incidence,* by race/ethnicity and U.S. census region --- United States, 1998--2006†。 Average annual rates of lung cancer diagnosed per 100,000 persons

http://www.cdc.gov/mmwr/preview/mmwrhtml/mm5944a2.htm



Drug delivery – Age/Area



http://www.cancerresearchuk.org/health-professional/**cancer-statistics**/mortality/age#heading-Zero

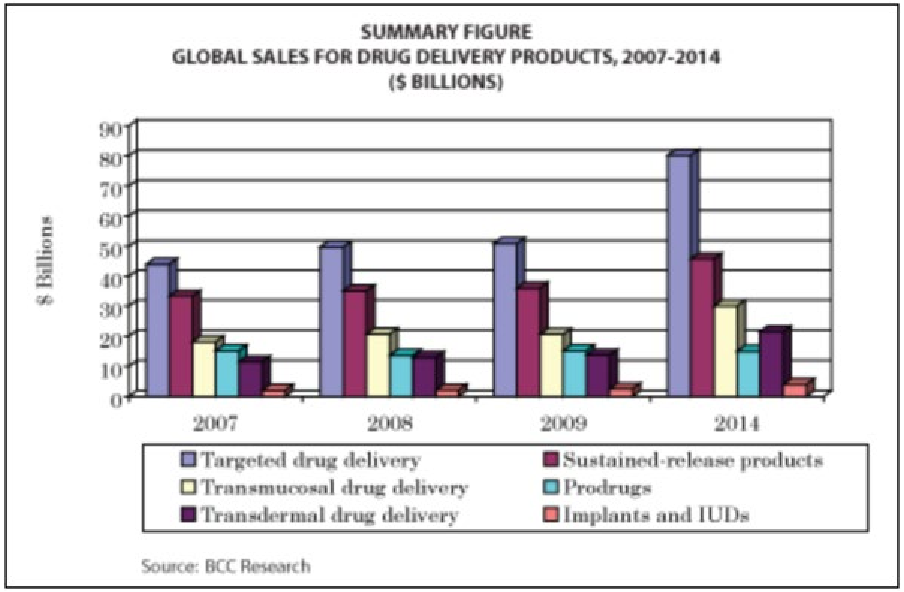
Global cancer rates

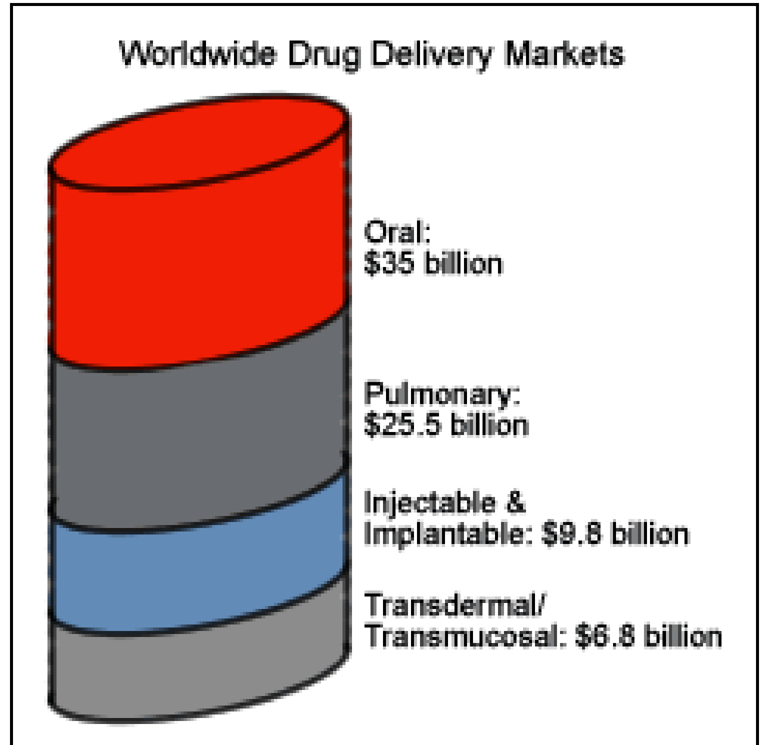
The ten worst countries and the UK cancer rate, per 100,000 of the population

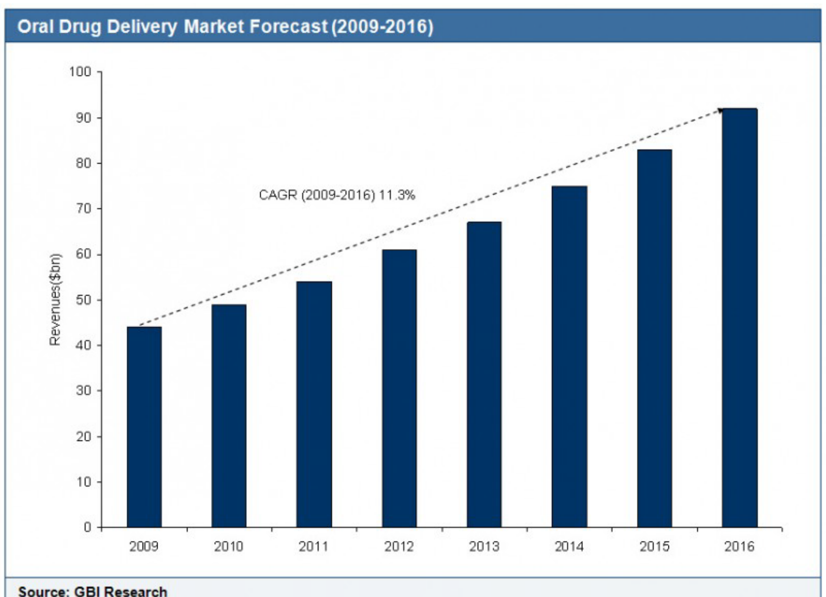
1	Denmark	326.1
2	Ireland	317
3	Australia	314.1
4	New Zealand	309.2
5	Belgium	306.8
6	France	300.4
7	US	300.2
8	Norway	299.1
9	Canada	296.6
10	Czech Republic	295
22	UK	266.9

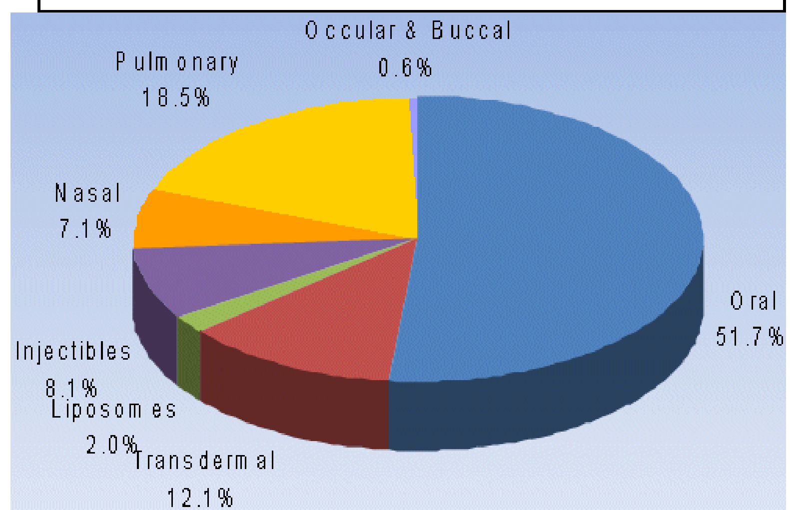
SOURCE: WORLD CANCER RESEARCH FUND

Drug delivery – Market

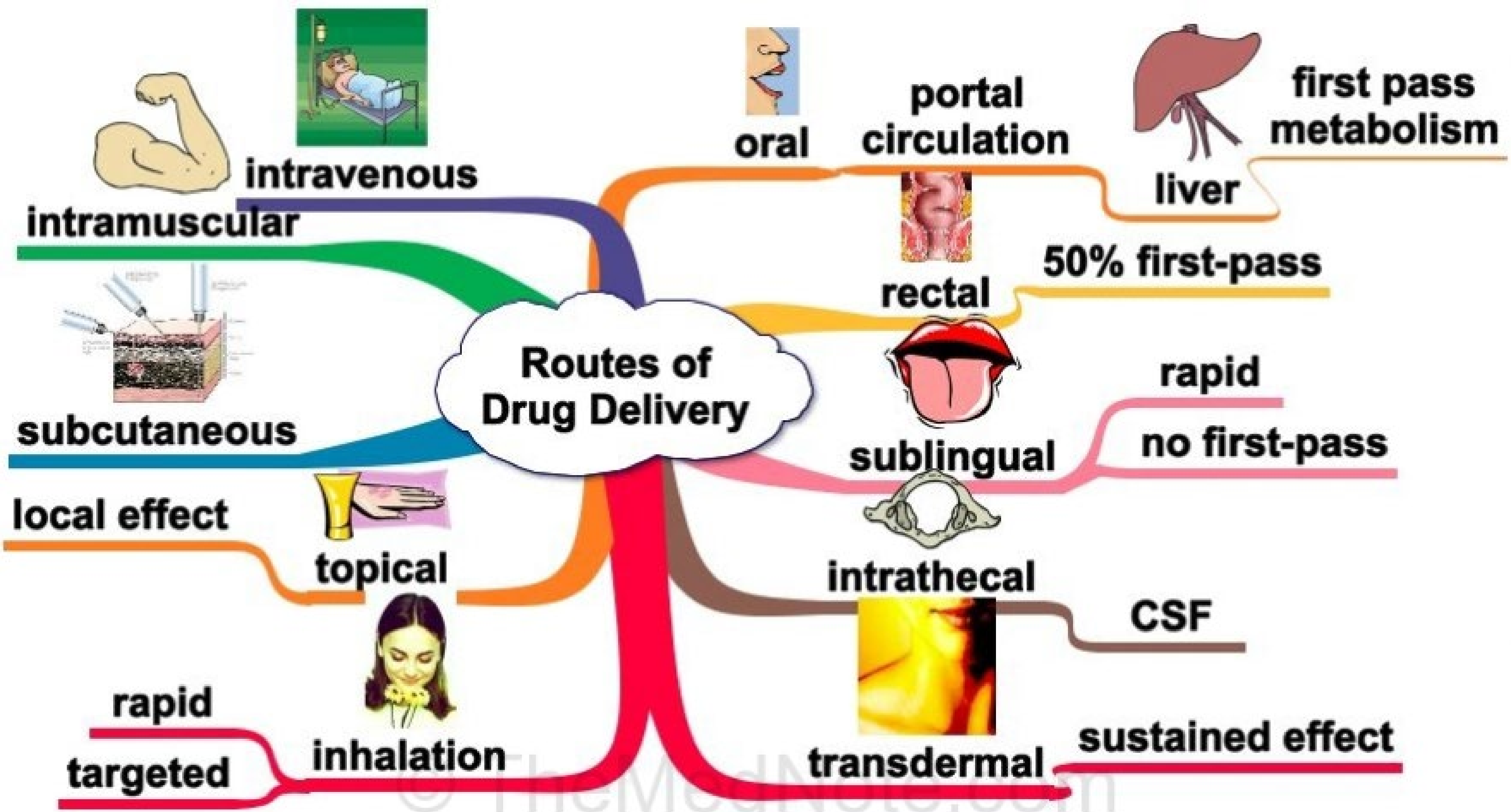








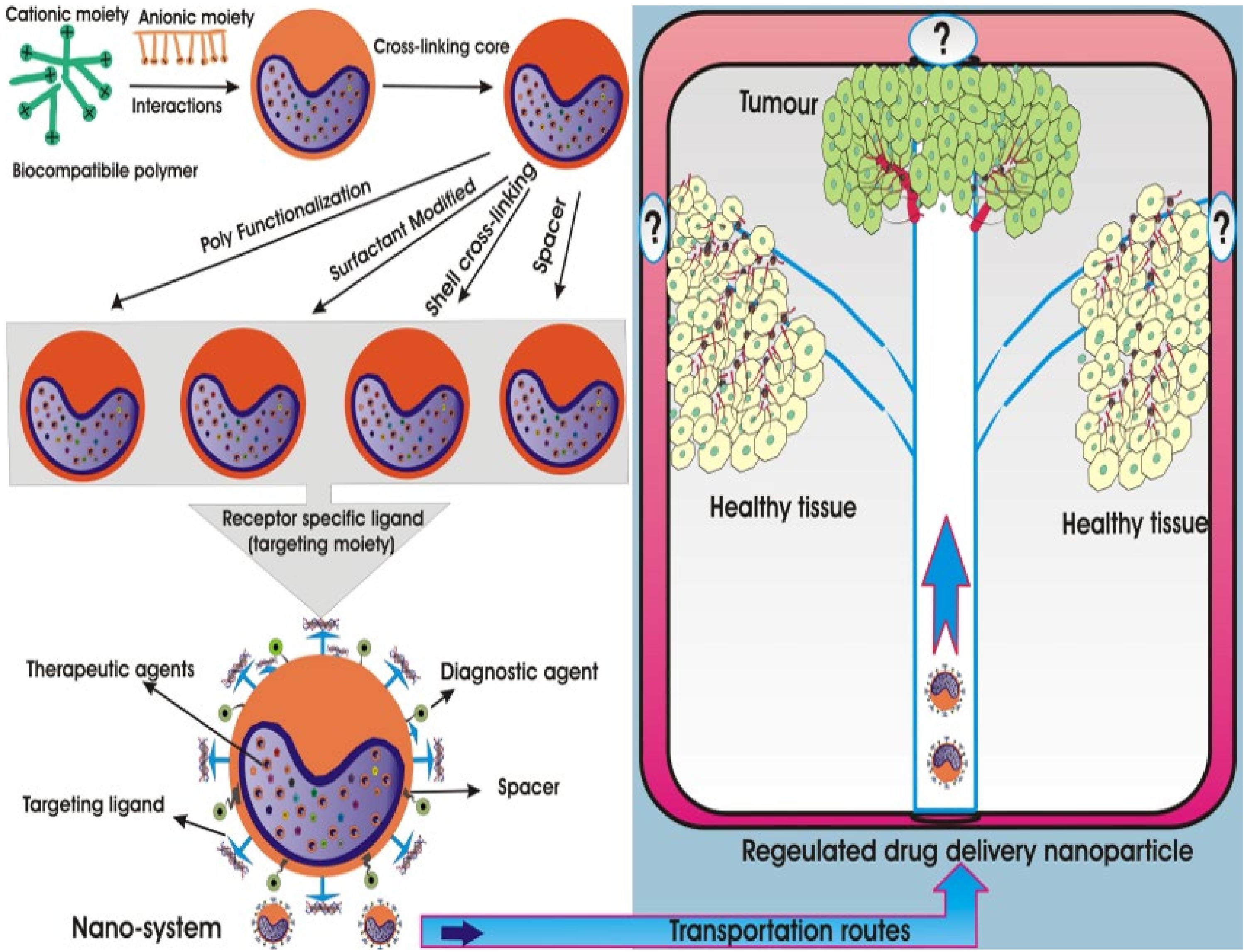
Drug delivery - Routes



http://www.themednote.com/2011/06/28/routes-of-drug-delivery/#.VxSbQ3oSmt4

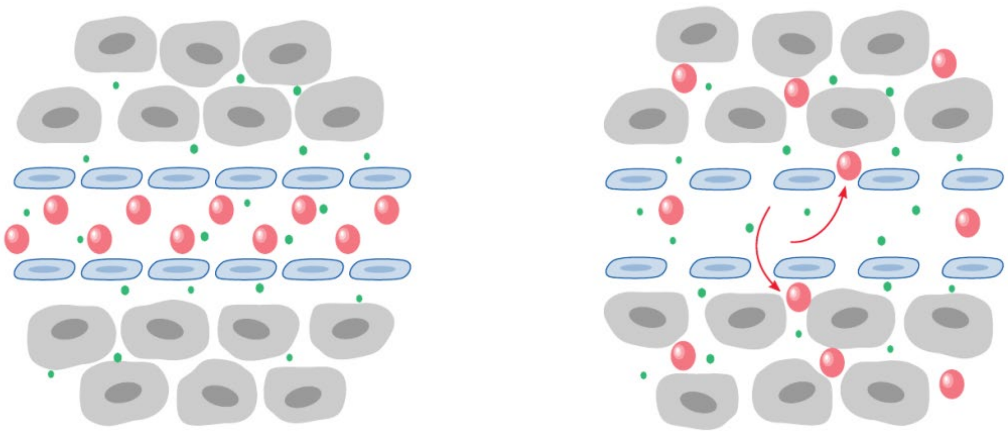
The first pass effect (also known as first-pass metabolism or presystemic metabolism) is a phenomenon of drug metabolism whereby the concentration of a drug is greatly reduced before it reaches the systemic circulation.

Drug delivery – Delivery Approaches



http://www.intechopen.com/books/application-of-nanotechnology-in-drug-delivery/polymer-nanoparticles-for-smart-drug-delivery

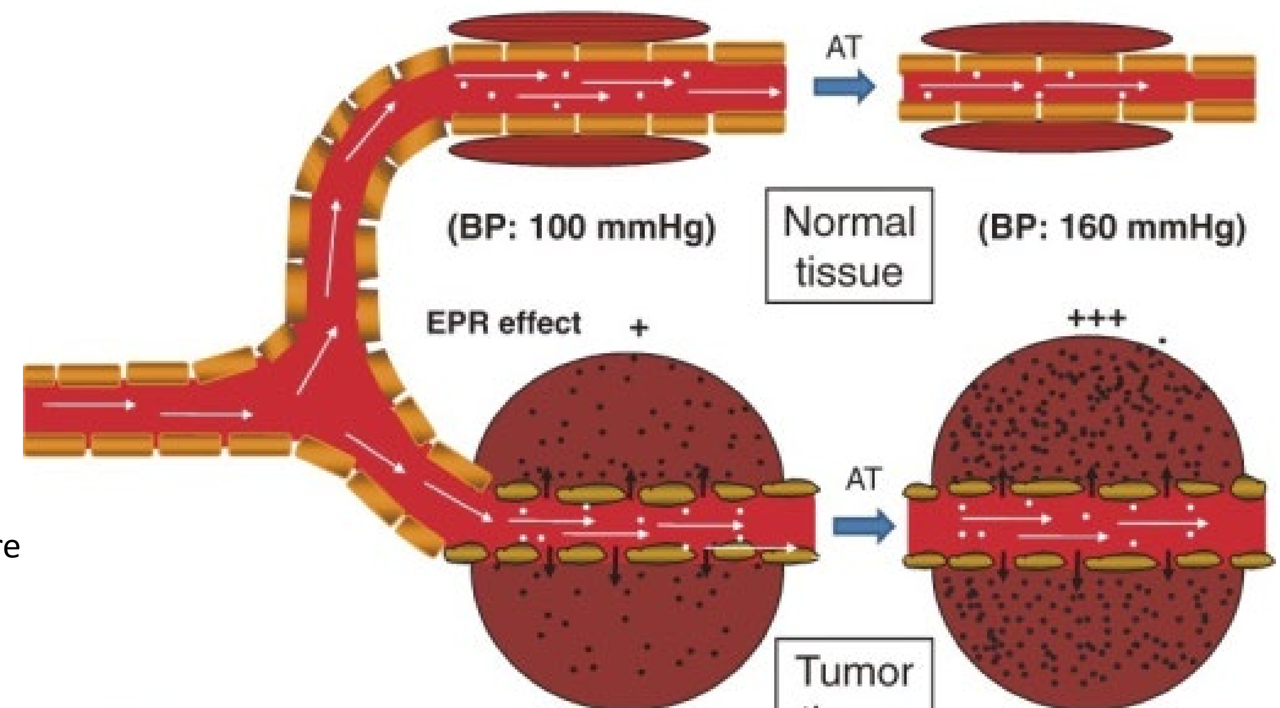
Drug delivery – EPR



(A) Normal vasculature: no leakage

EPR

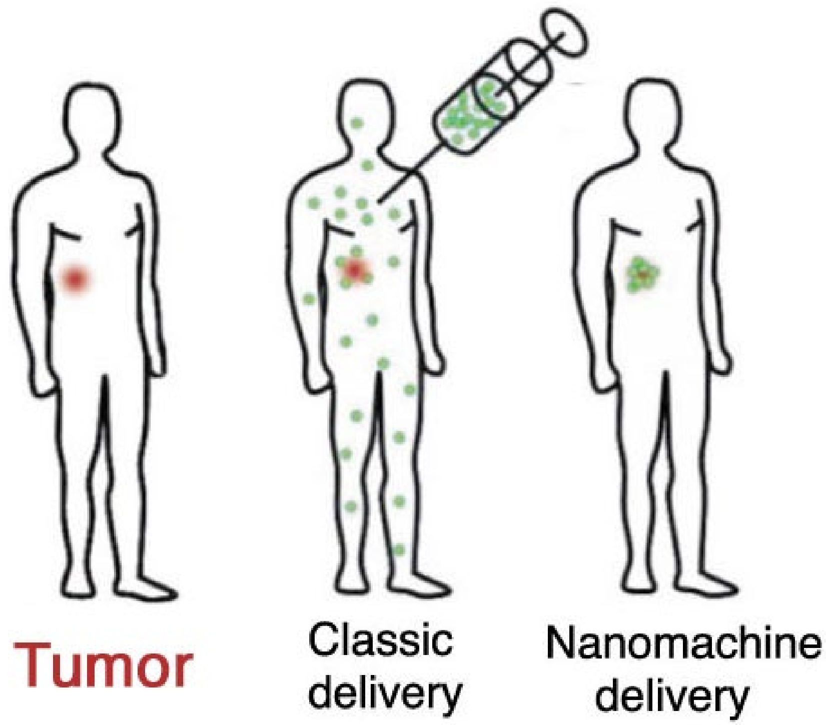
Enhanced permeability and retention effect



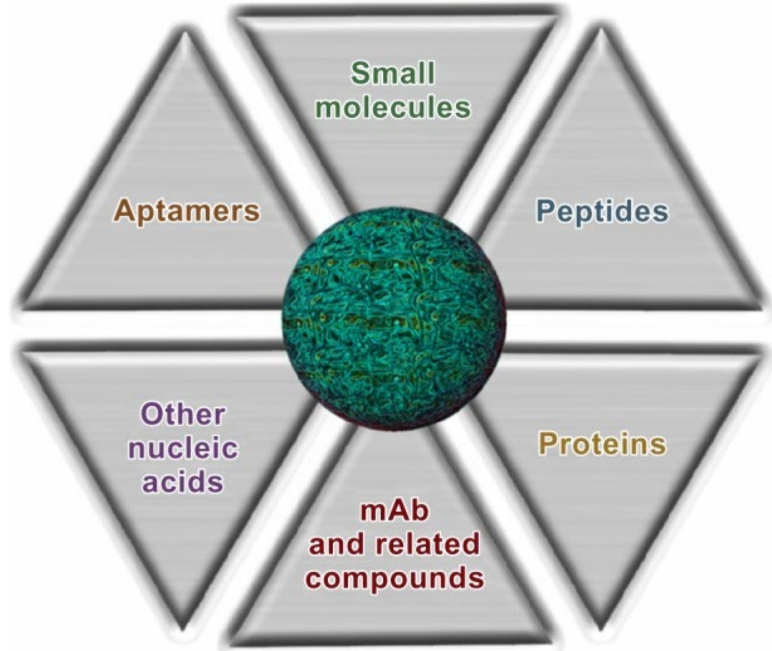
(B) Tumor vasculature: leakage

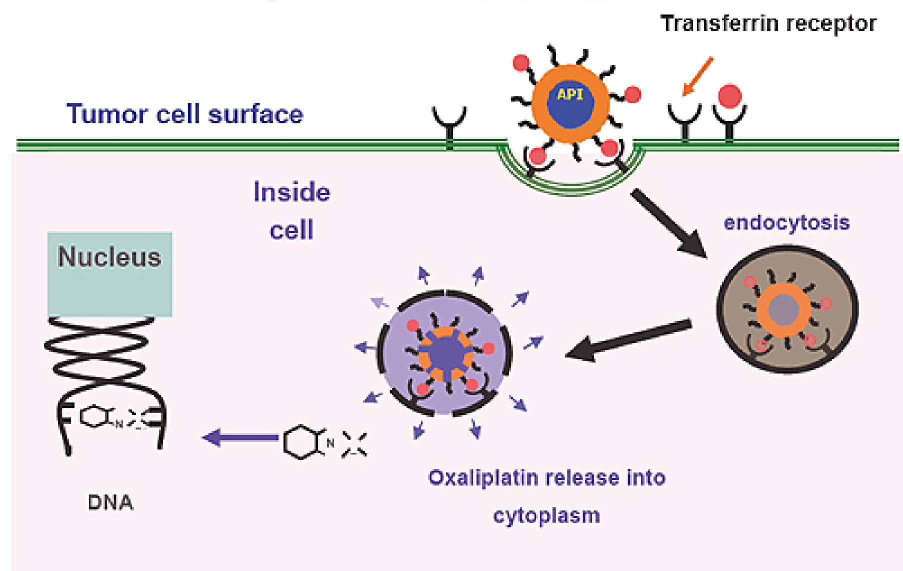
https://openi.nlm.nih.gov/detailedresult.php?img=PMC3365245_pjab-88-053-g008&req=4

Drug delivery Nontarget/Target delivery



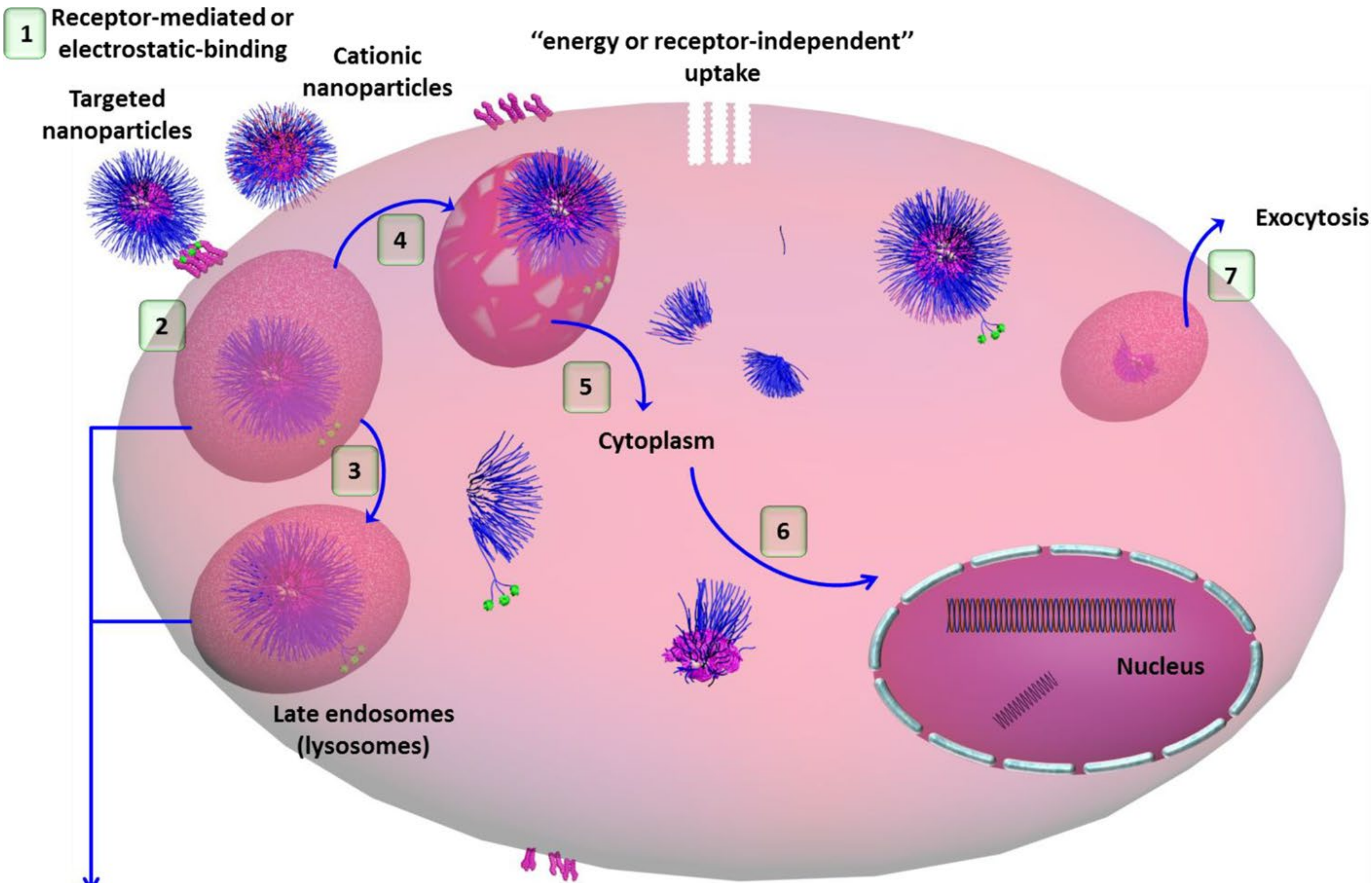
http://www.nanomachineslab.org/portofolio/drug_delivery/





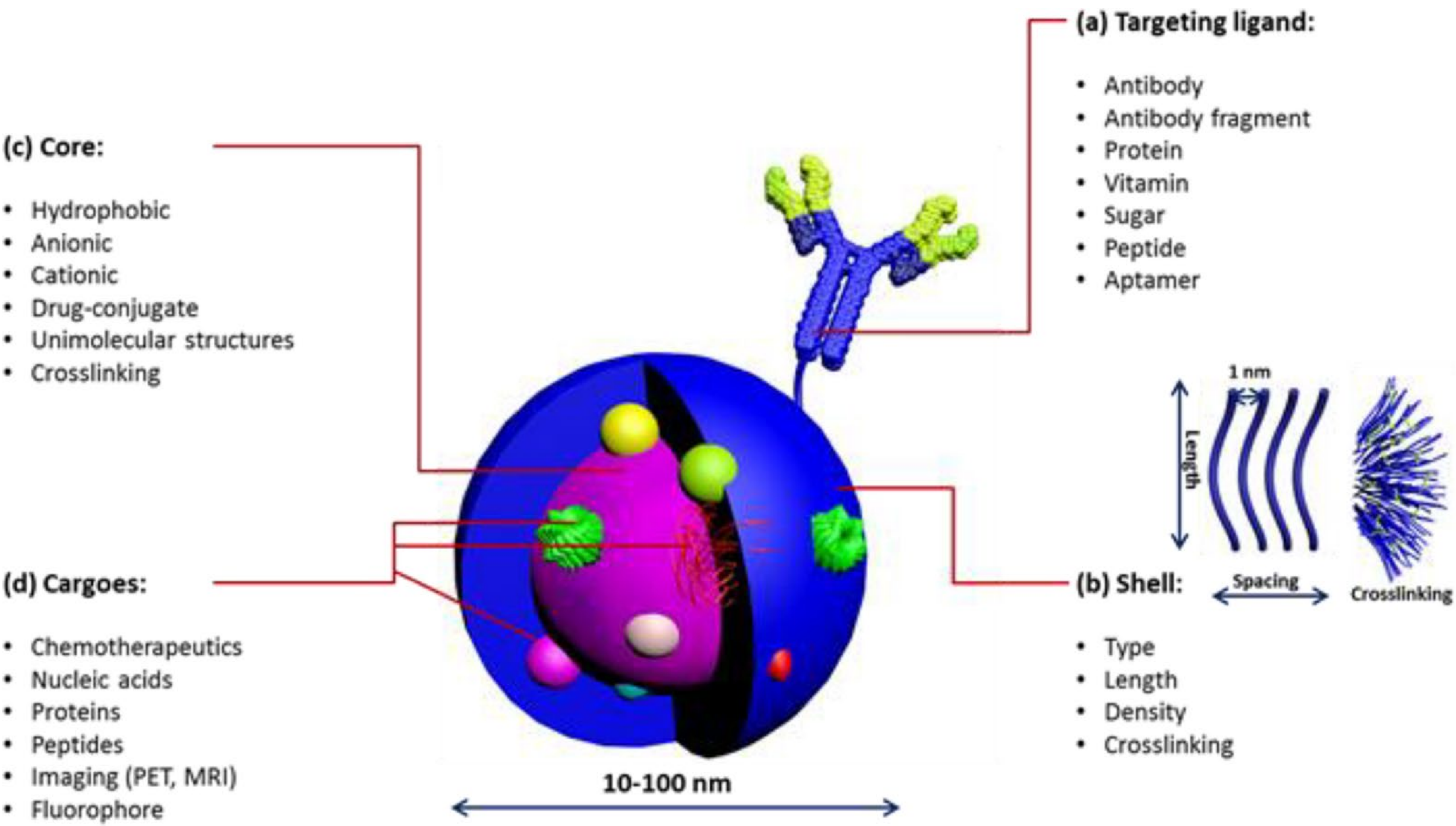
http://www.mebiopharm.com/english/pro.html

Drug delivery

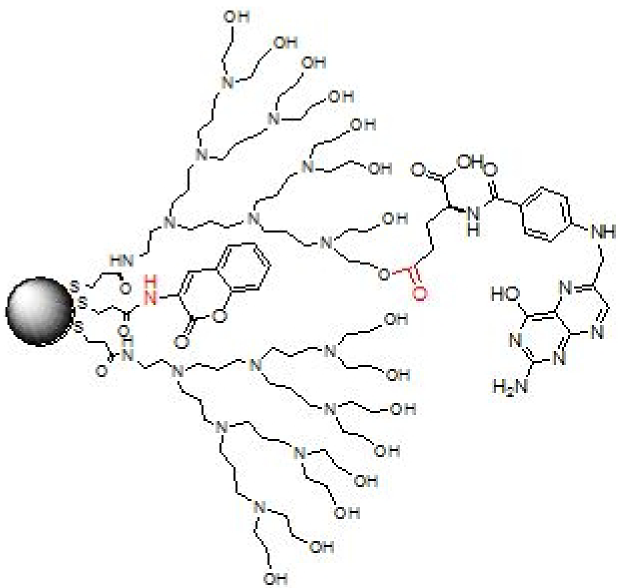


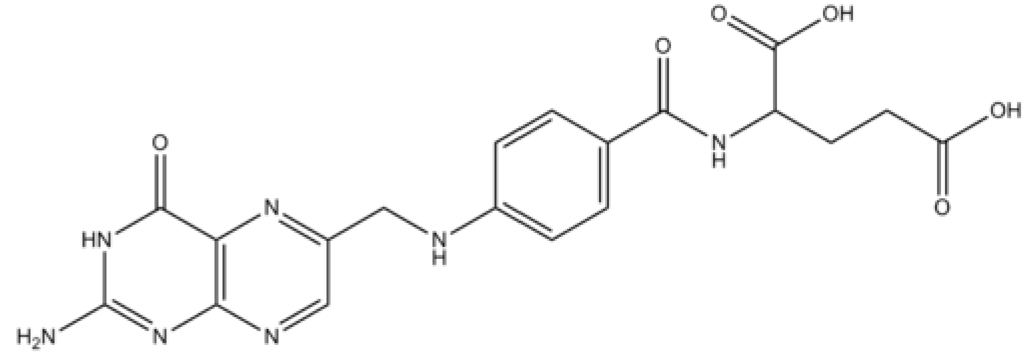
Depending on the endocytic pathway (phagocytosis, macropinocytosis, clathrin-mediated, caveolin-mediated, or clathrin/caveolin-independent endocytosis) and the proteins assist in the endocytic process, vesicles of various compositions and sizes (*e.g.*, endosomes, phagosomes, macropinosomes) are usually formed

Drug delivery

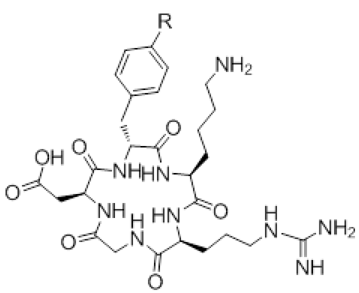


Drug delivery

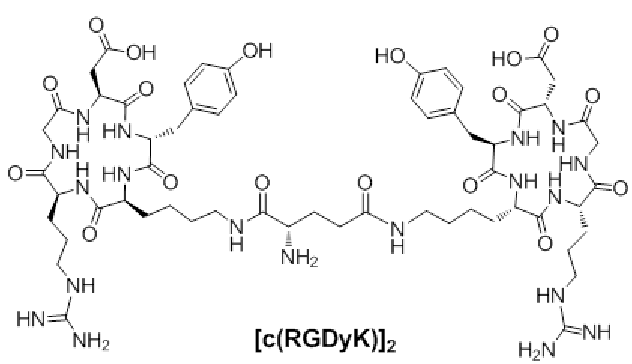




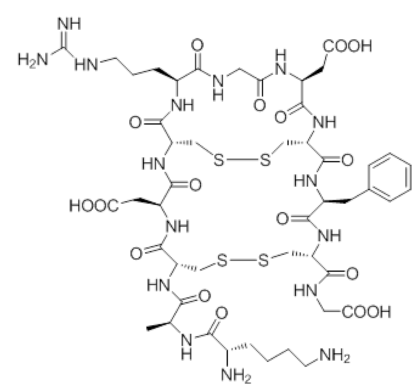
Folic acid



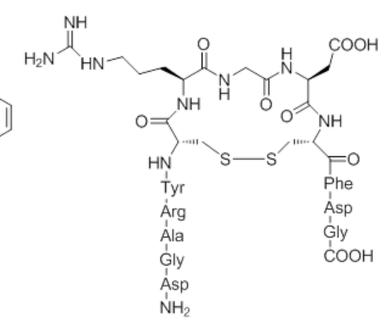
c(RGDyK) (R = OH)
c(RGDfK) (R = H)



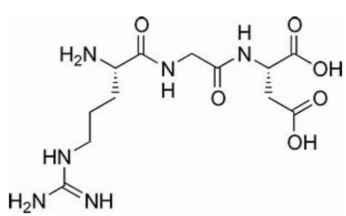
[c(RGDyK)]$_2$



RGD4C
(ACDC**RGD**CFCG)

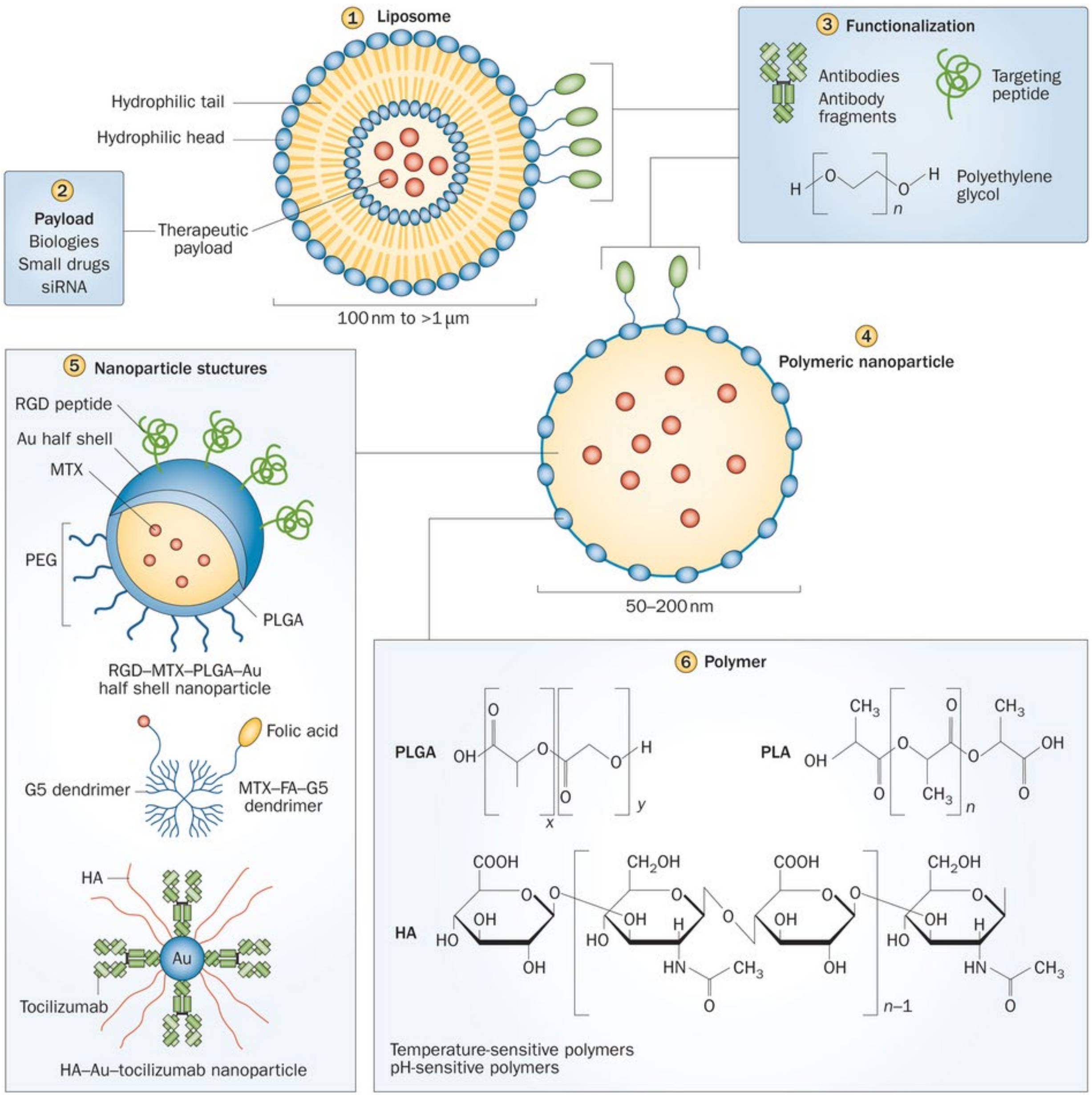


RGD10
(DGARYC**RGD**CFDG)



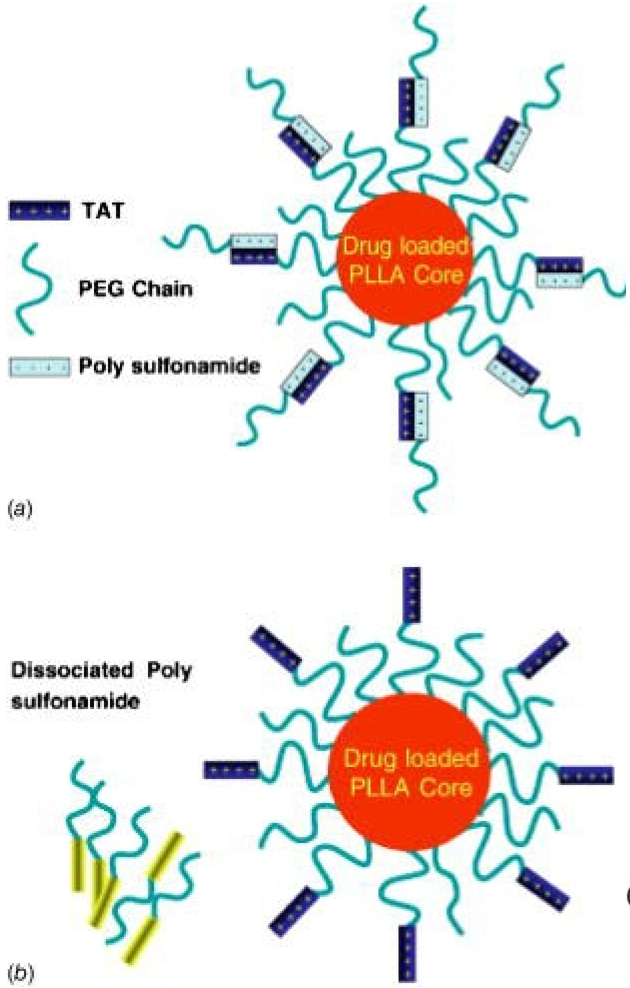
RGD: Arginine-Glycine-Aspartic Acid

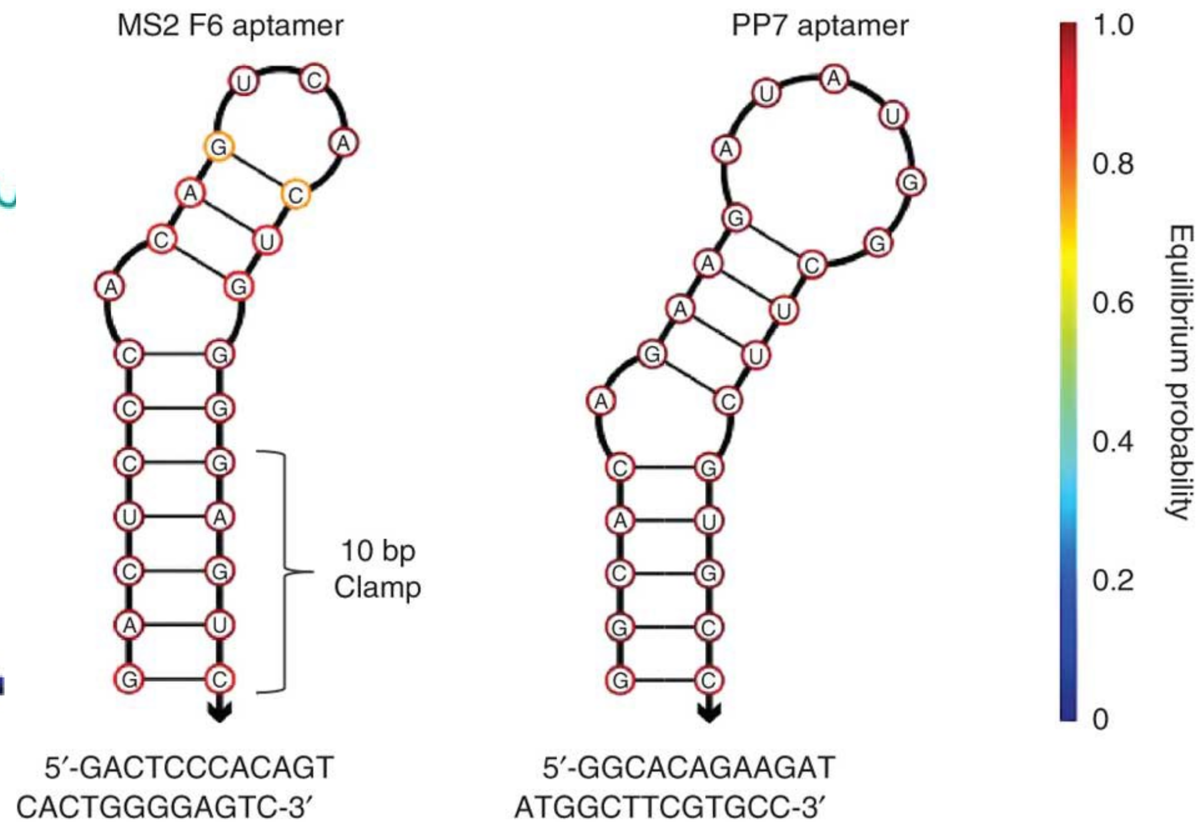
Drug delivery



Nature Reviews | Rheumatology

Drug delivery

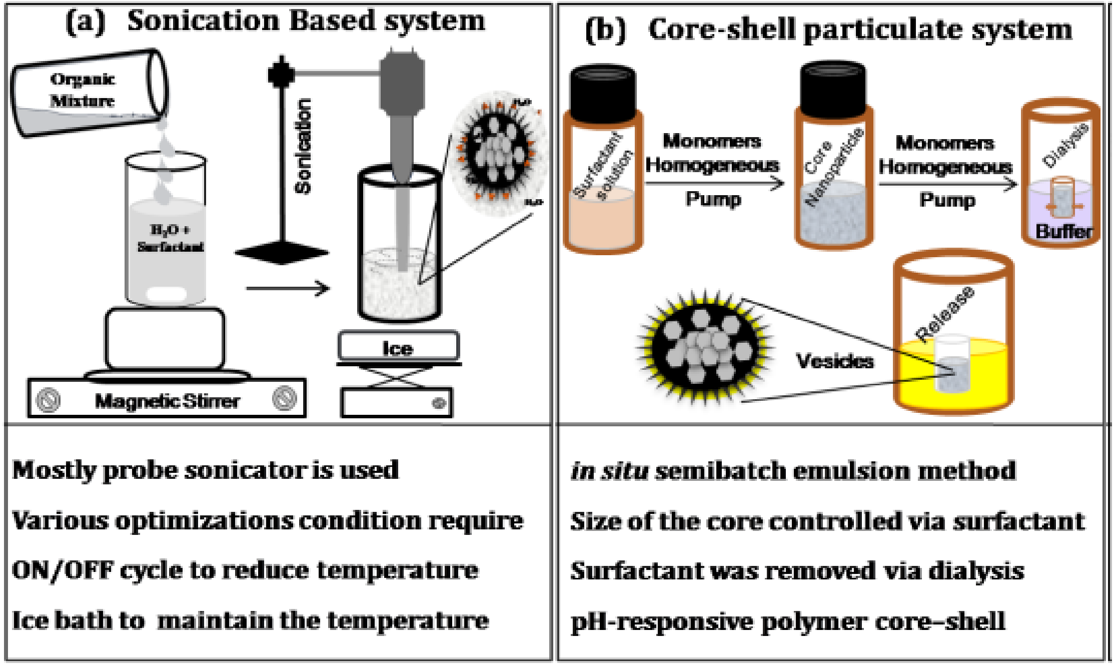




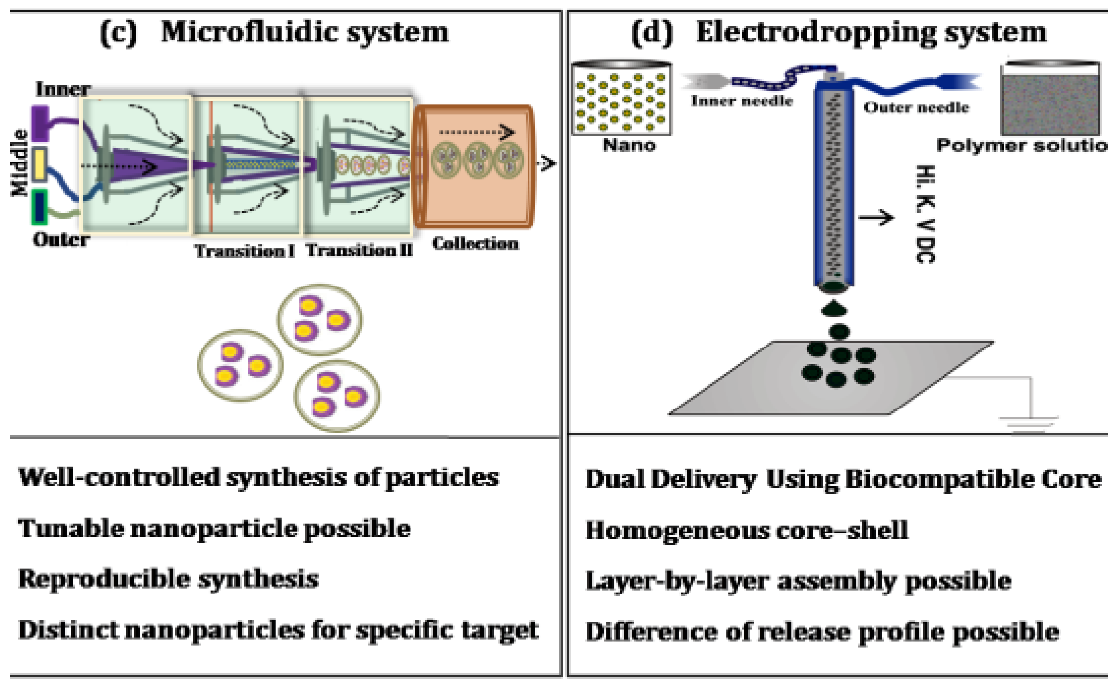
TATpeptide——YGRKKRRQRRRC

TAT peptide 是一种可渗透细胞的肽 (GRKKRRQRRRPQ),来自HIV-1反转录激活因子 (Tat)

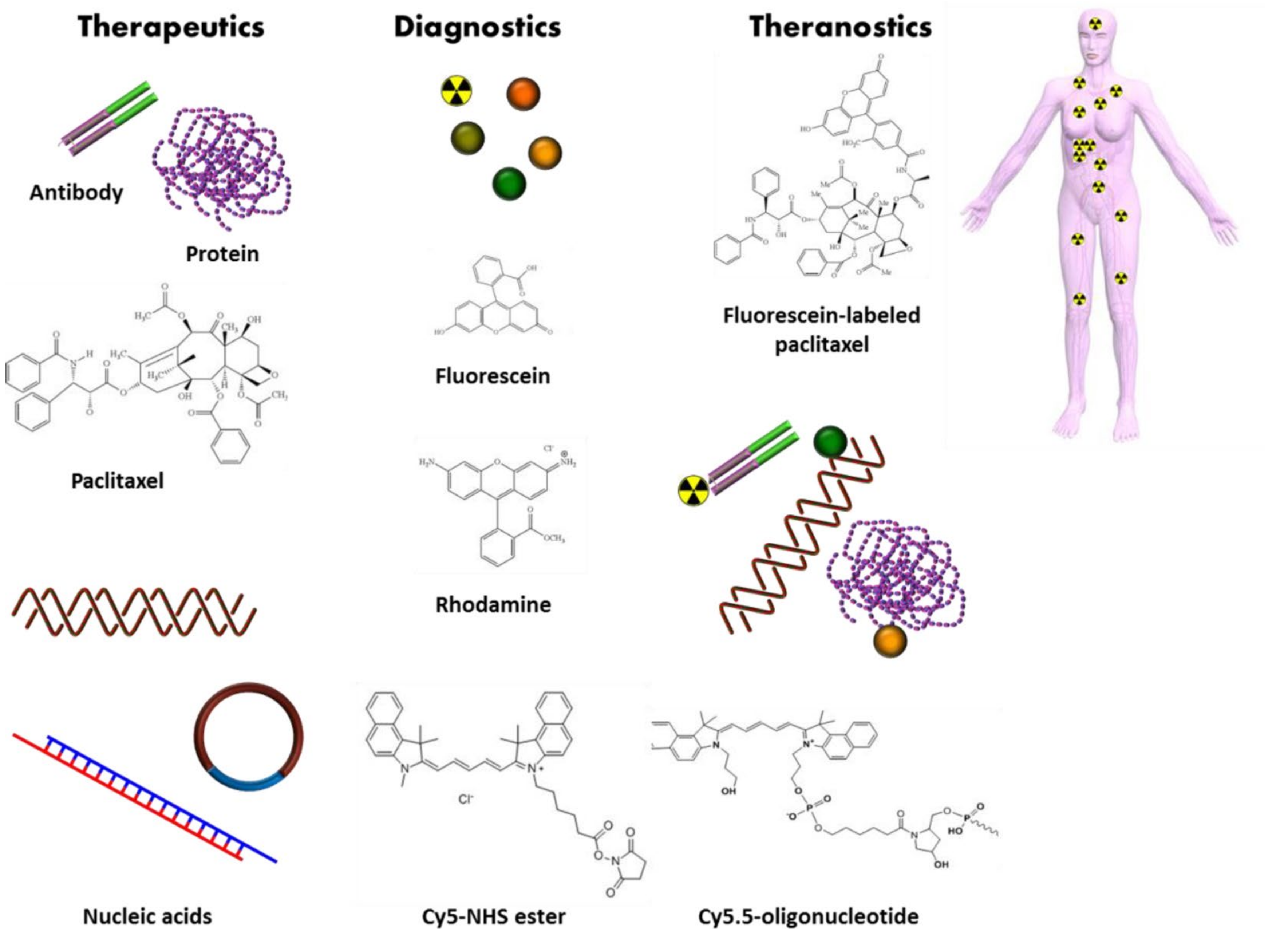
核酸适配体（Aptamer）是一段寡核苷酸序列（DNA或RNA）



http://cdn.intechopen.com/pdfs-wm/46807.pdf



Drug delivery



Chem. Rev. 2015, 115, 10967−11011

Drug delivery

Formulation

- Low solubility
- Hydrolytic or enzymatic degradability
- Premature release
- Instability

Administration

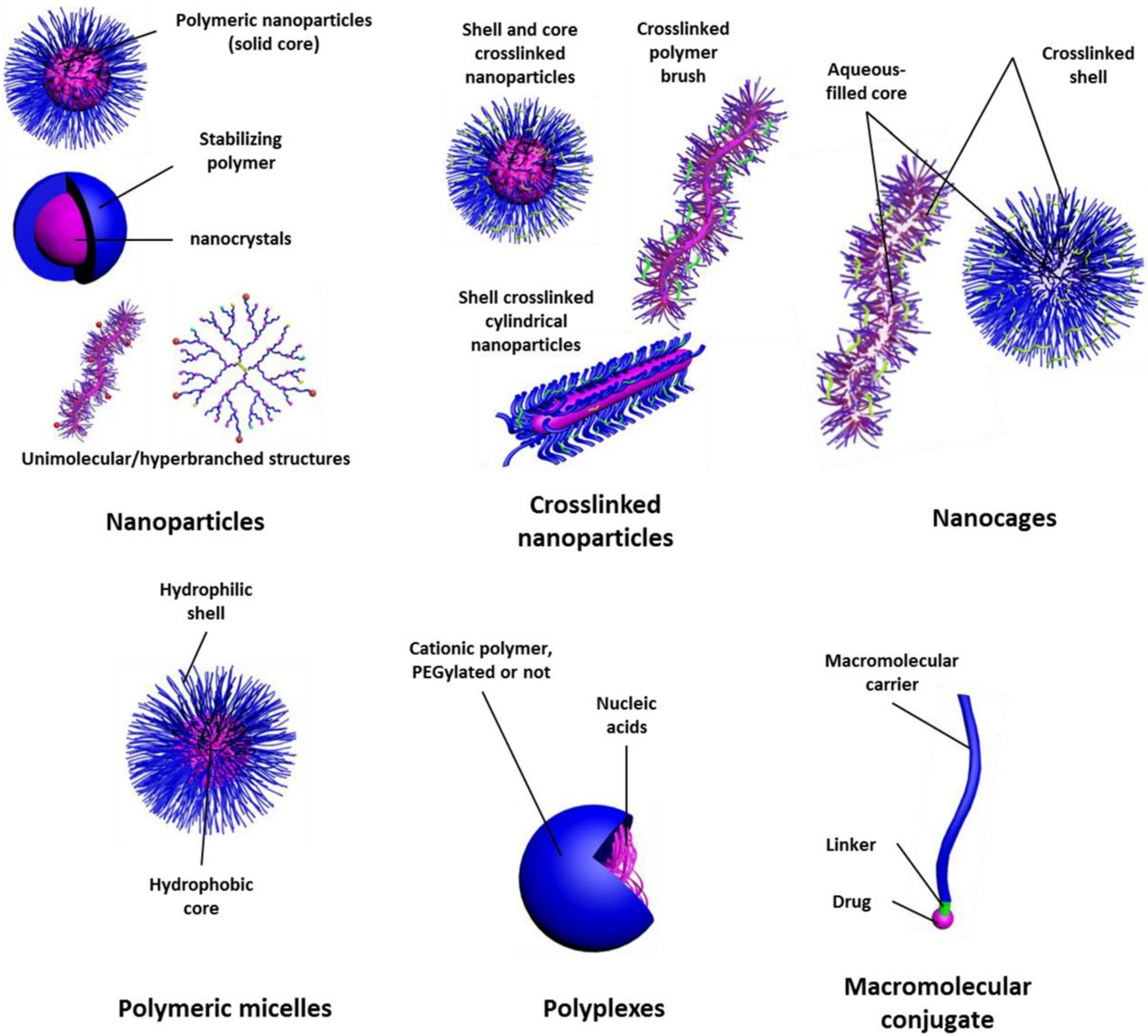
- Low penetration through skin or mucus
- Toxicity
- Opsonization
- Destabilization
- Rapid clearance (renal, hepatic, splenic)
- Rapid degradation

Biodistribution

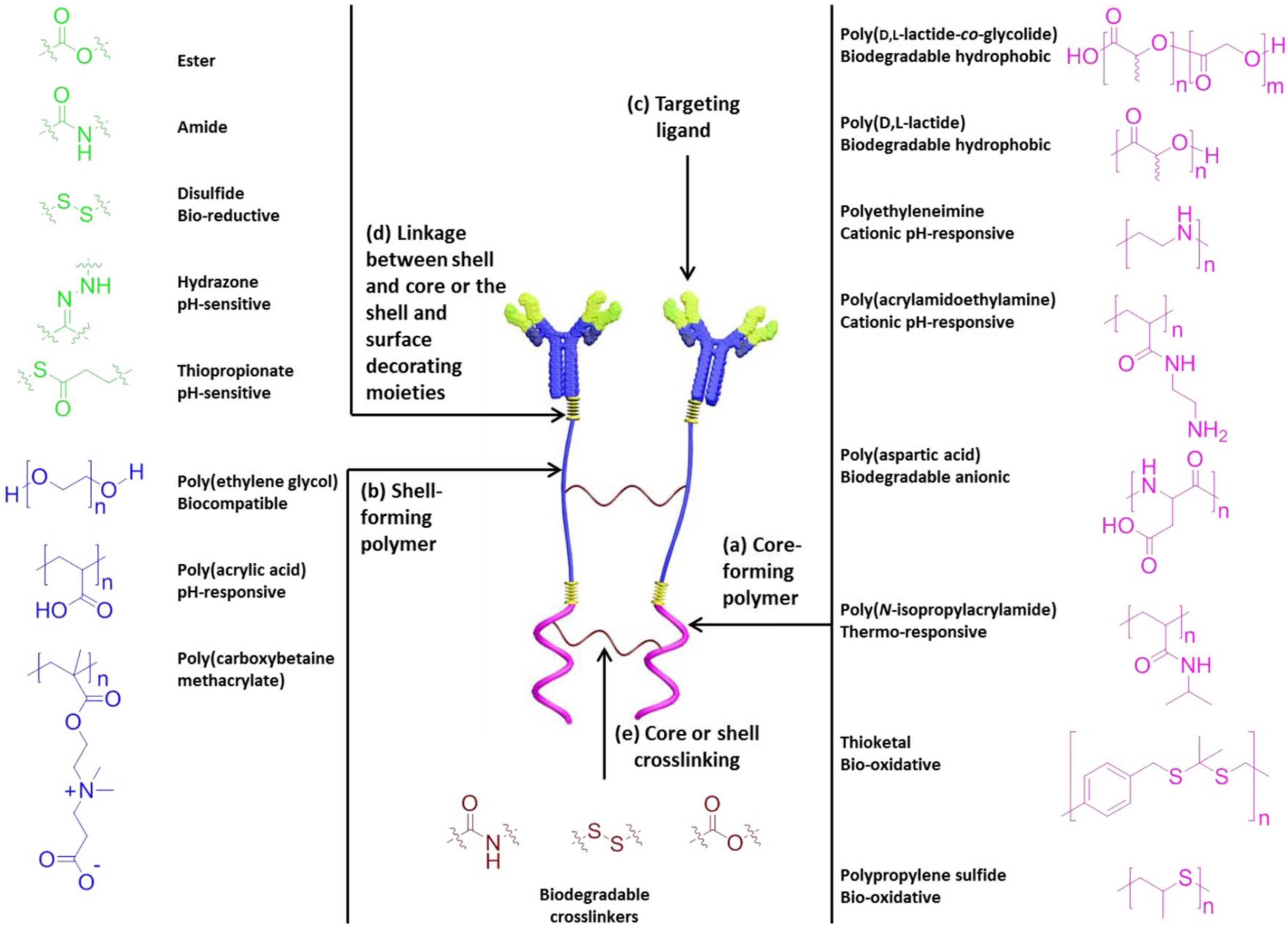
- Poor selectivity
- Non-specific biodistribution
- Low intracellular bioavailability

Main challenges toward the use of various therapeutics, diagnostics, and theranostics during formulation, administration, and the resulting biodistribution, with no control over their characteristics and navigation within the biological system.

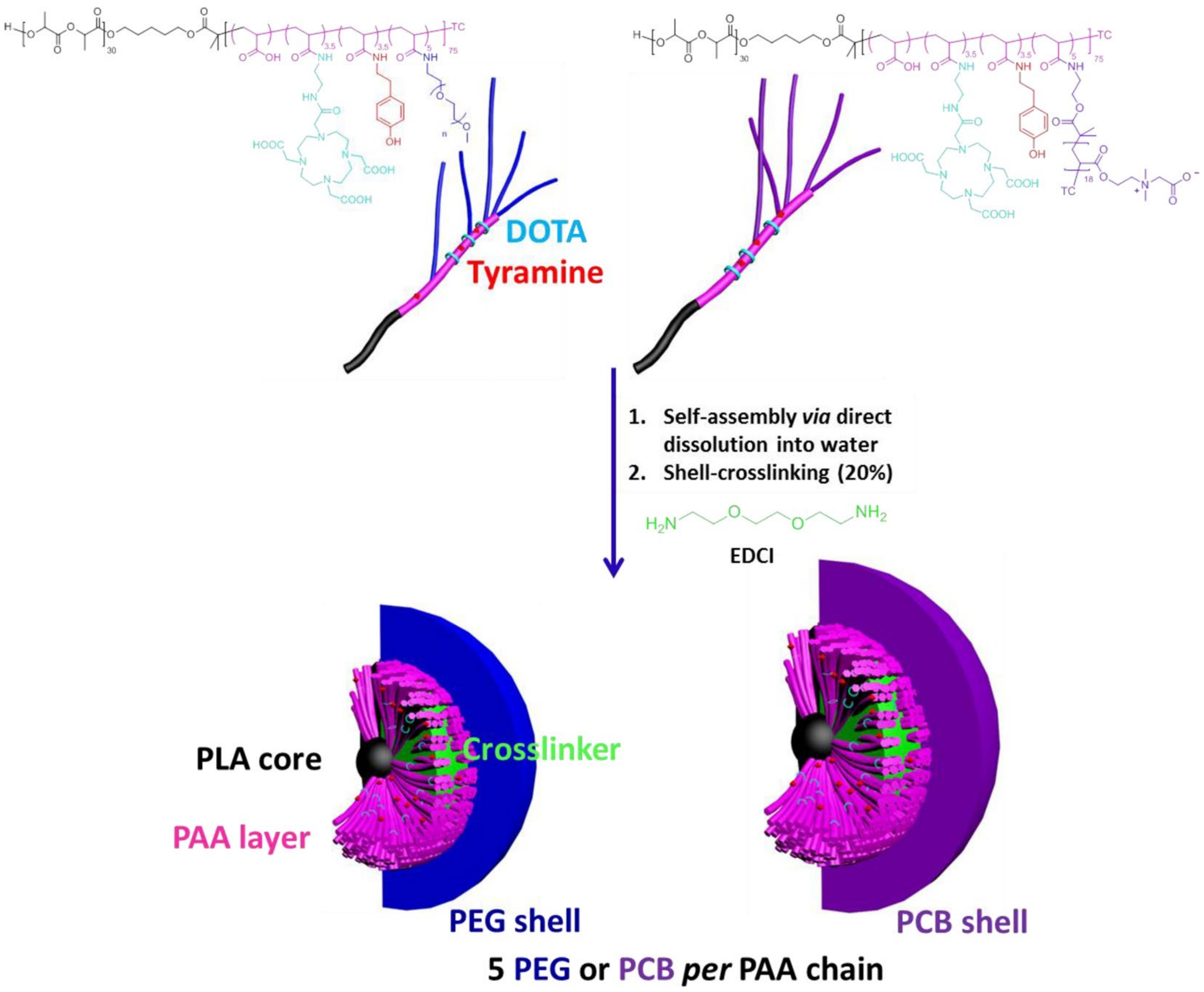
Drug delivery



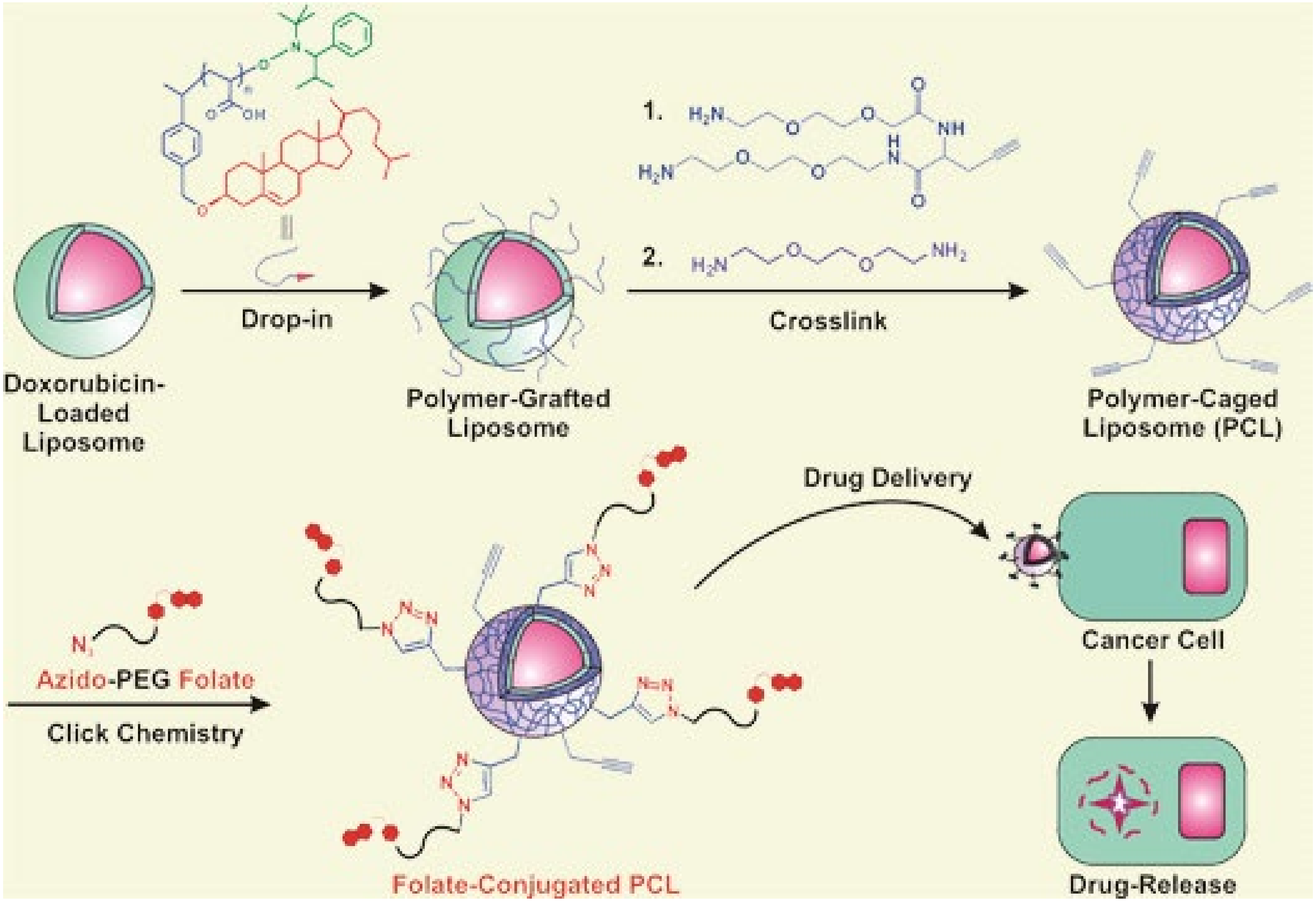
Drug delivery



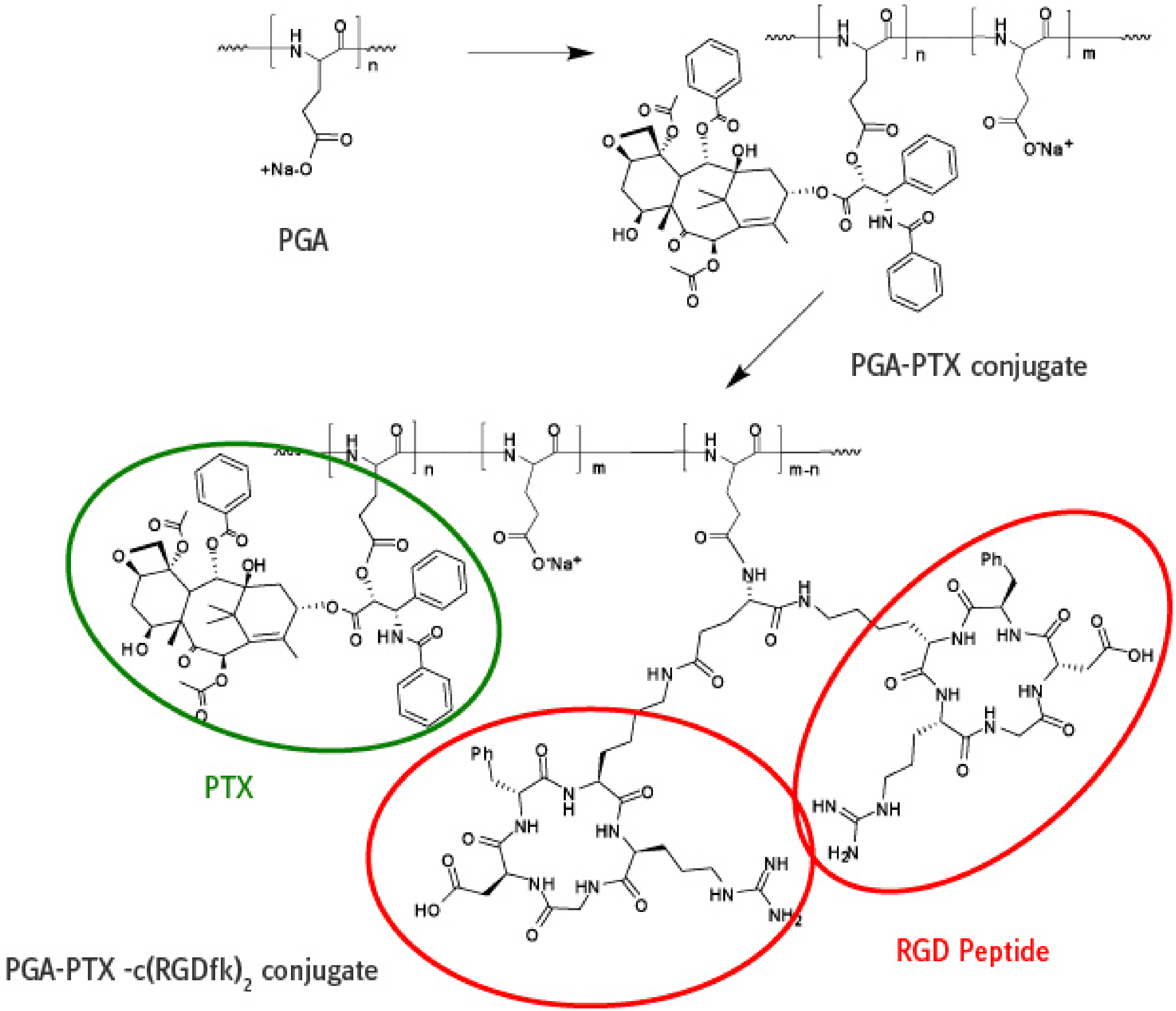
Drug delivery



Drug delivery



Drug delivery



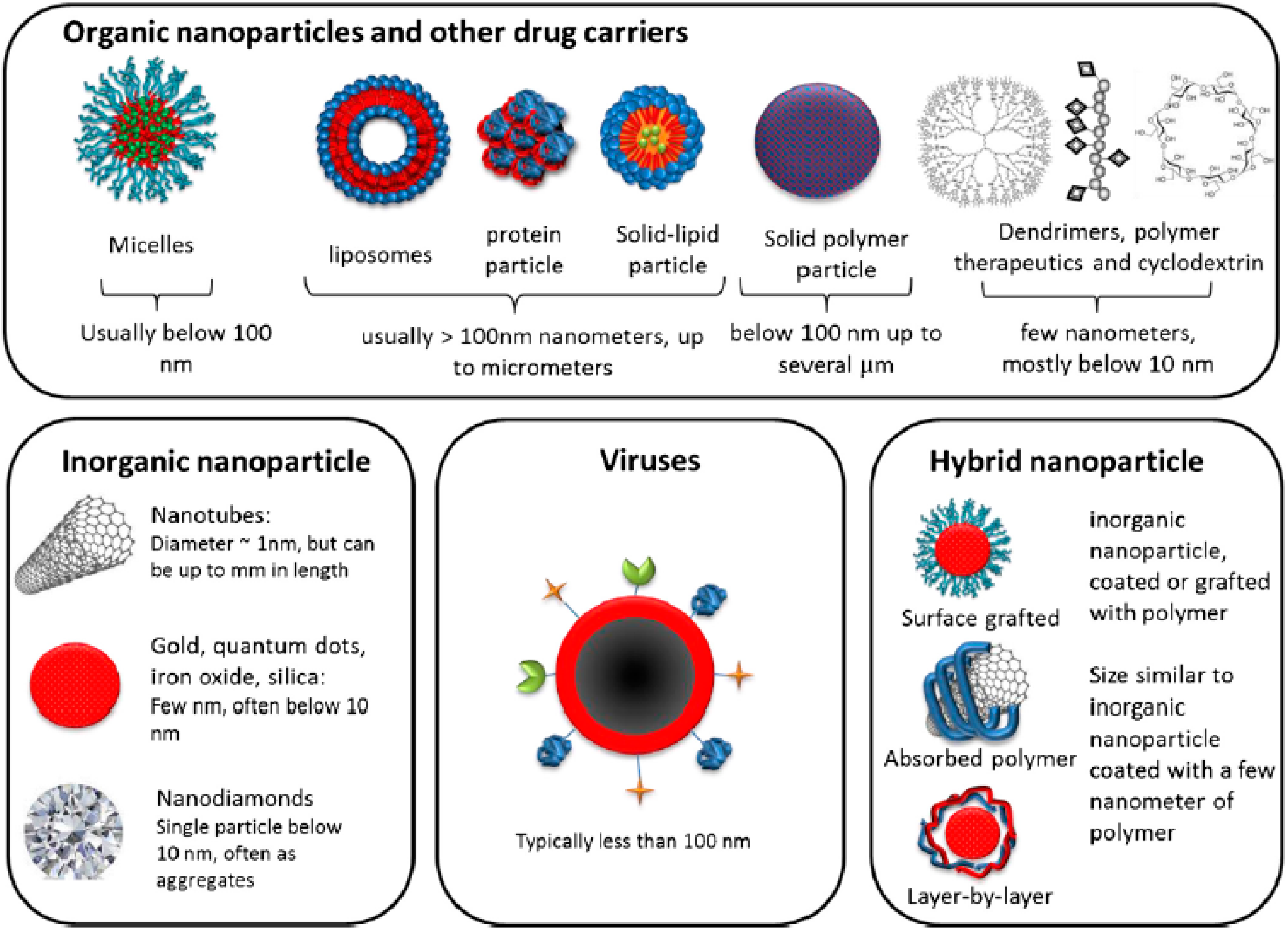
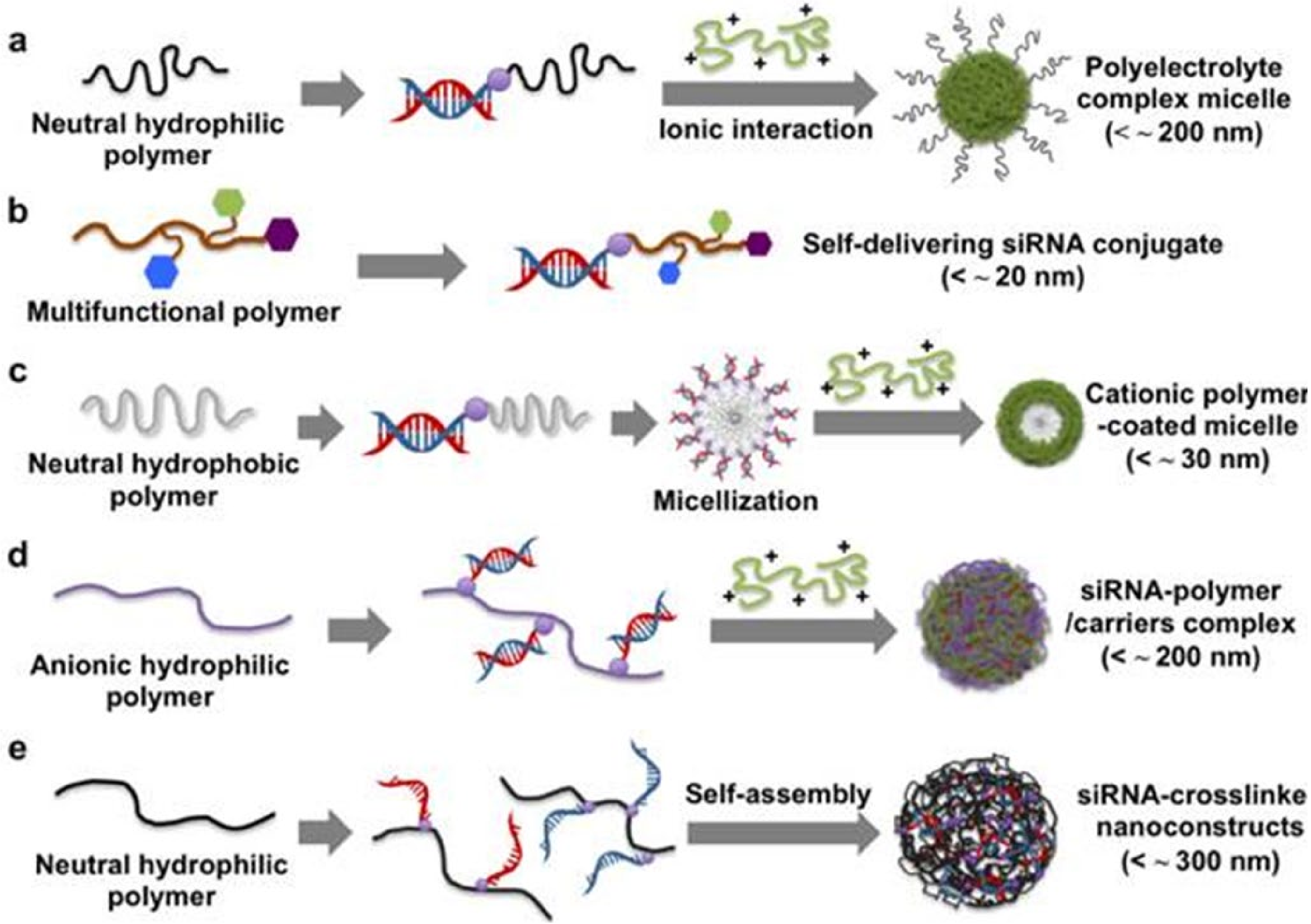


Figure 4. Nanoparticle illustrations that exist in both soft and hard (Reproduced with permission from [1]. Copyright Elsevier, 2016).

https://www.mdpi.com/2079-4991/7/4/77

siRNA Delivery



http://www.thno.org/v04/p1211/thnov04p1211g002.jpg

siRNA Delivery

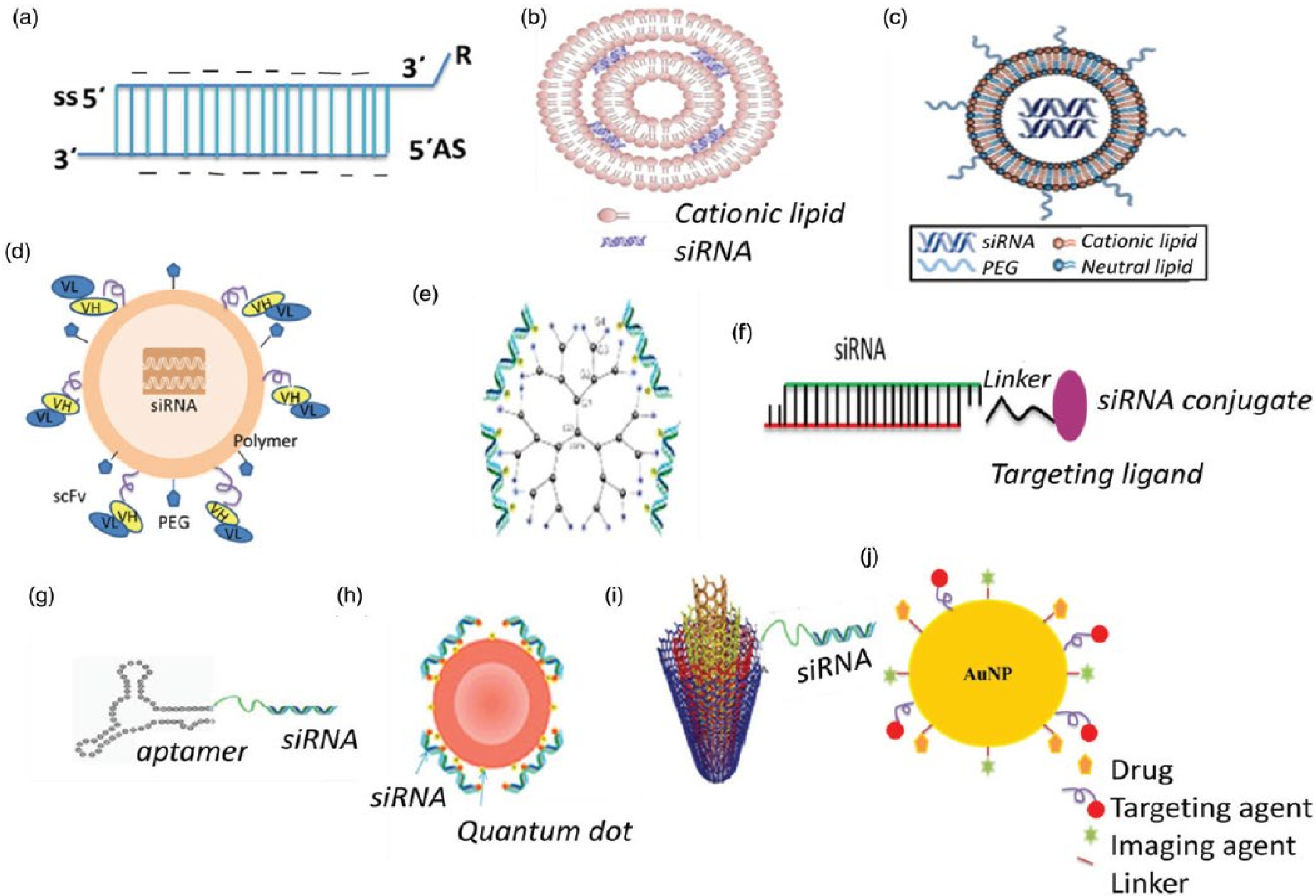
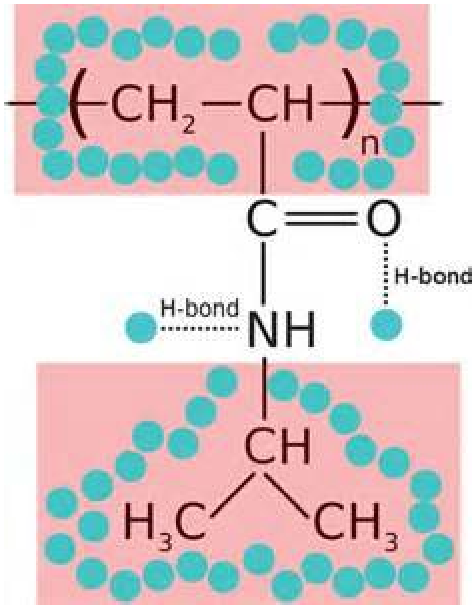
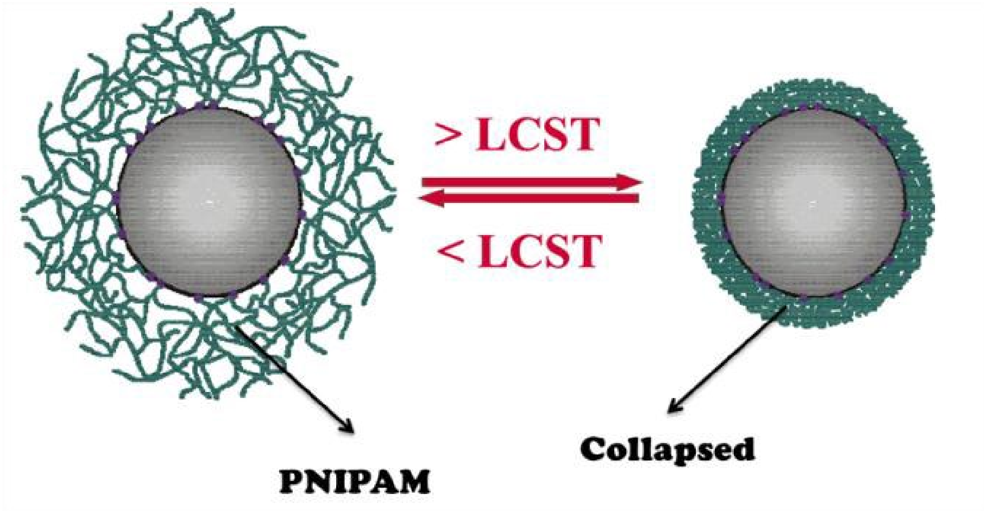


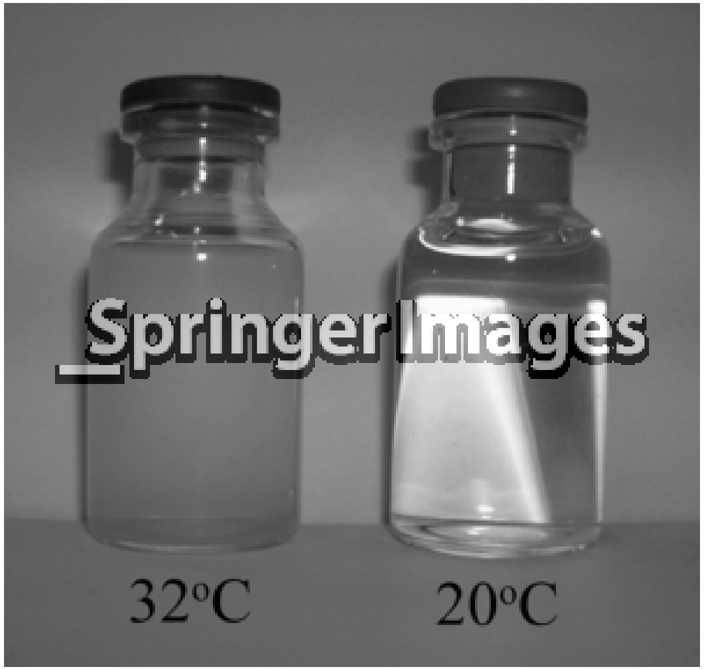
Figure 4. Various types of siRNA cancer therapeutics: (a) siRNA strand chemical modification, (b) Lipoplexes, (c) stable nucleic acid-lipid particles (SNALPs), (d) polymer-based delivery, (e) dendrimer-siRNA conjugate, (f) conjugate siRNA delivery, (g) aptamer-siRNA chimaeras, (h) quantum dot complexed with siRNA, (i) carbon nanotube-siRNA conjugate, (j) gold nanoparticles.

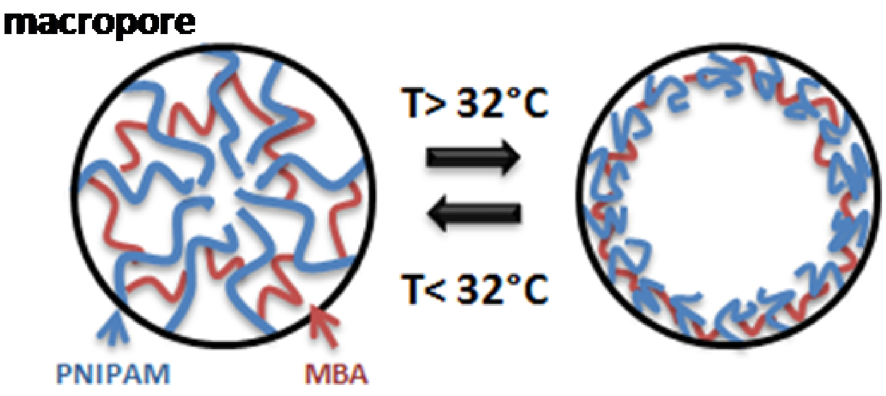
https://www.tandfonline.com/doi/pdf/10.1080/21691401.2017.1307210?needAccess=true

Drug delivery – Temperature sensitive

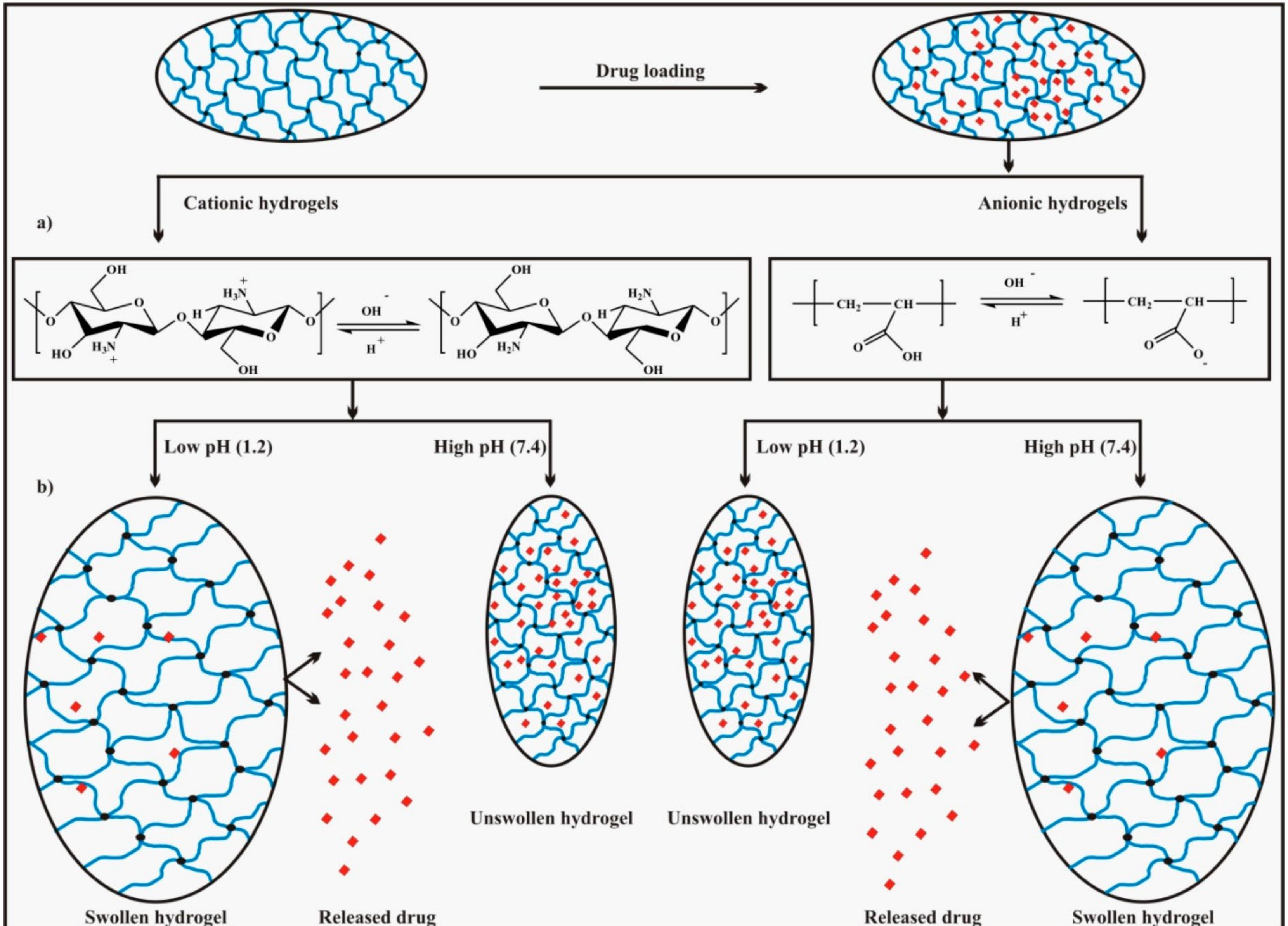








Drug delivery – pH sensitive



http://www.mdpi.com/2073-4360/9/4/137/htm

Drug delivery – pH sensitive

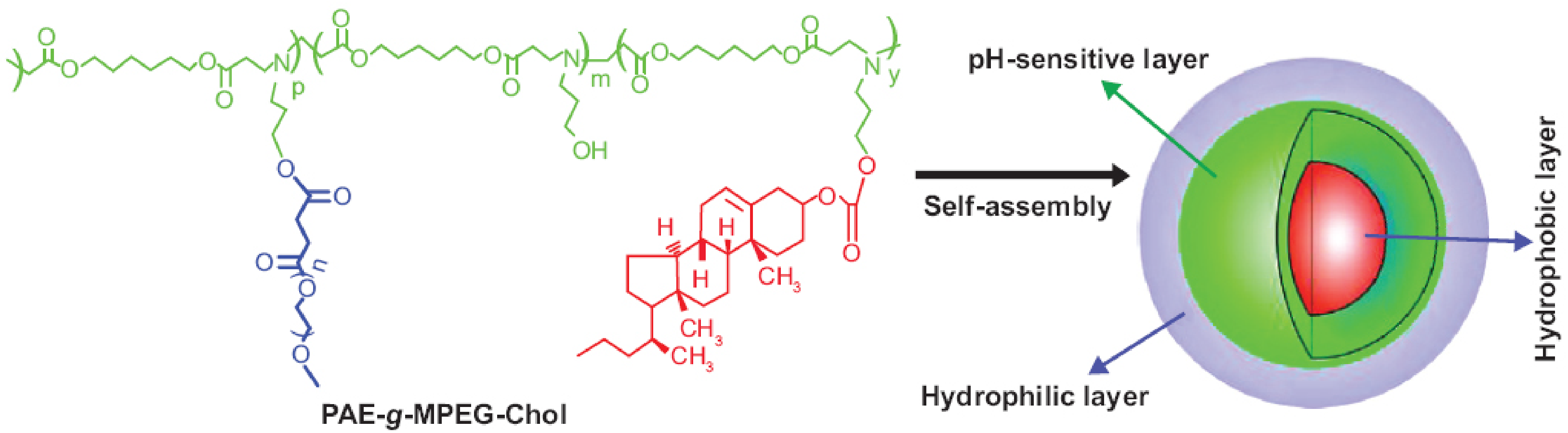
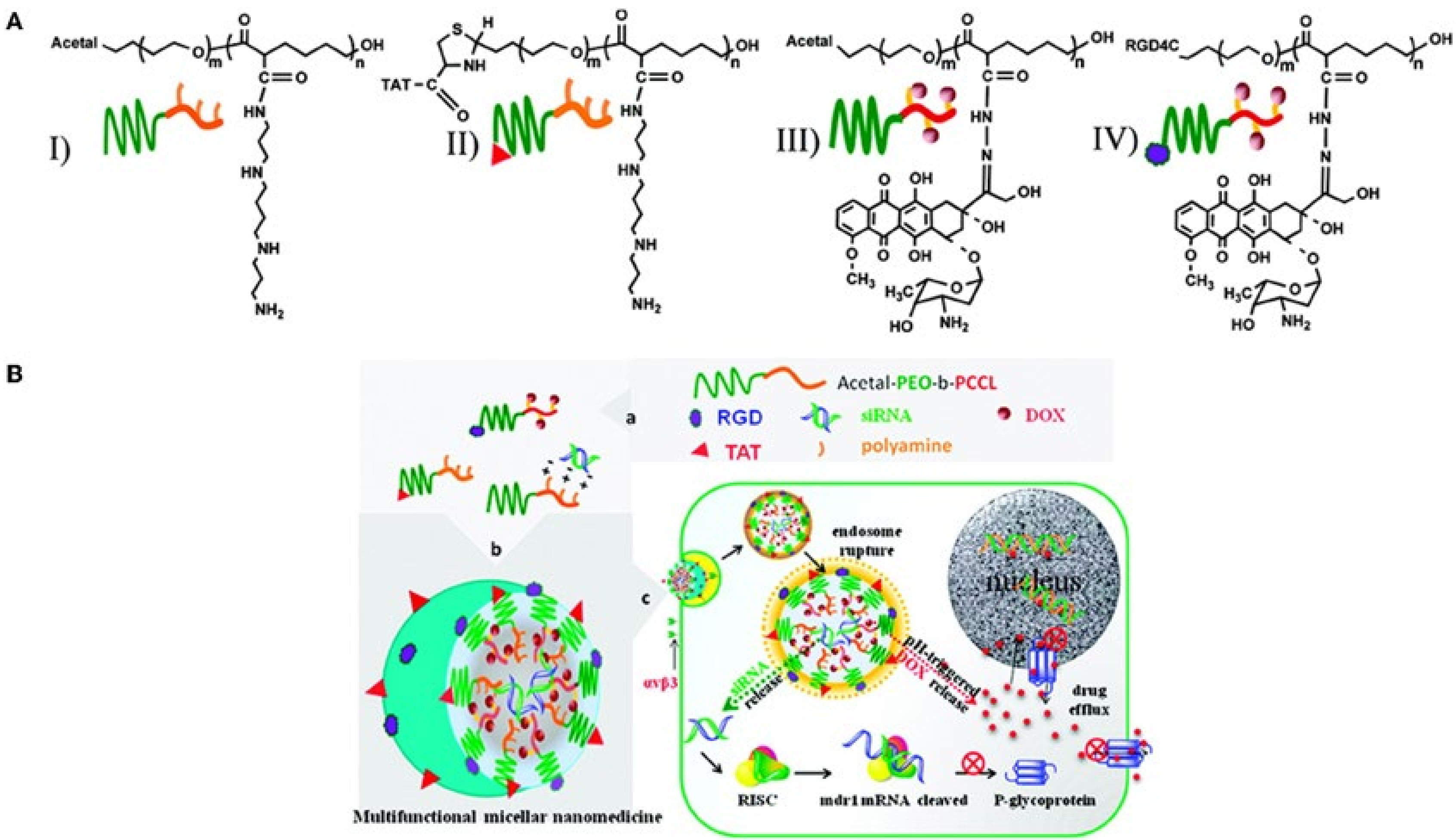


Figure 1 Schematic representation of micellization of poly(β-amino ester)-g-poly(ethylene glycol) methyl ether-cholesterol (PAE-g-MPEG-Chol).

https://www.dovepress.com/self-assembled-micelles-based-on-ph-sensitive-pae-g-mpeg-cholesterol-b-peer-reviewed-fulltext-article-IJN

Drug delivery – pH sensitive



https://www.frontiersin.org/articles/10.3389/fphar.2014.00077/full

Drug delivery – pH sensitive

Catechol Polymers for Cancer Drug Delivery

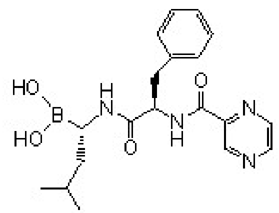
J. Su, et al., *JACS* 2011, 133, 11850.

Velcade® in Cancer Therapy:

- Proteasome inhibitor
- Approved for treatment of multiple myeloma

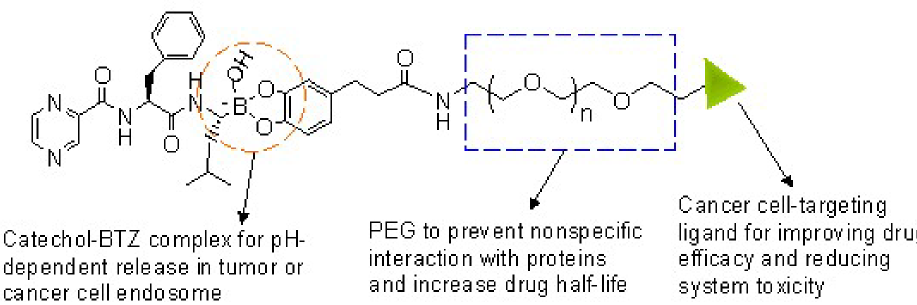
Limitations:

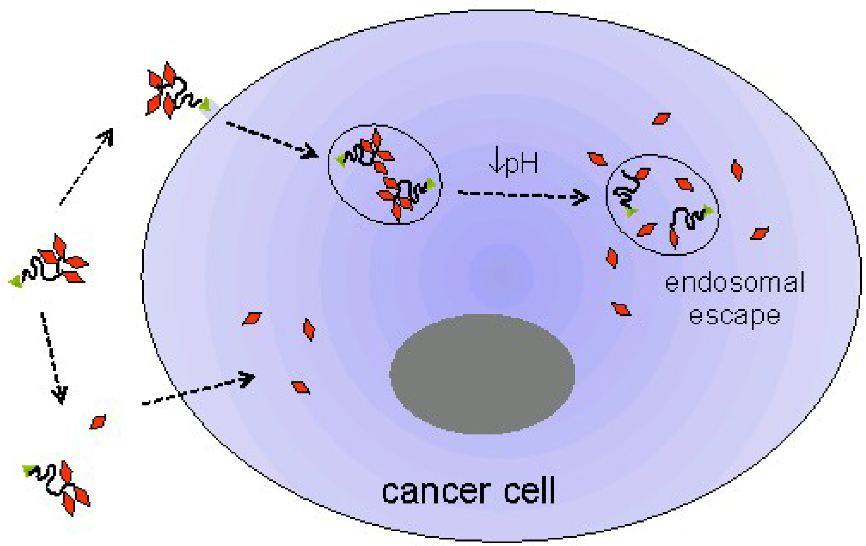
- Rapid clearance following IV administration
- Side effects including peripheral neuropathy

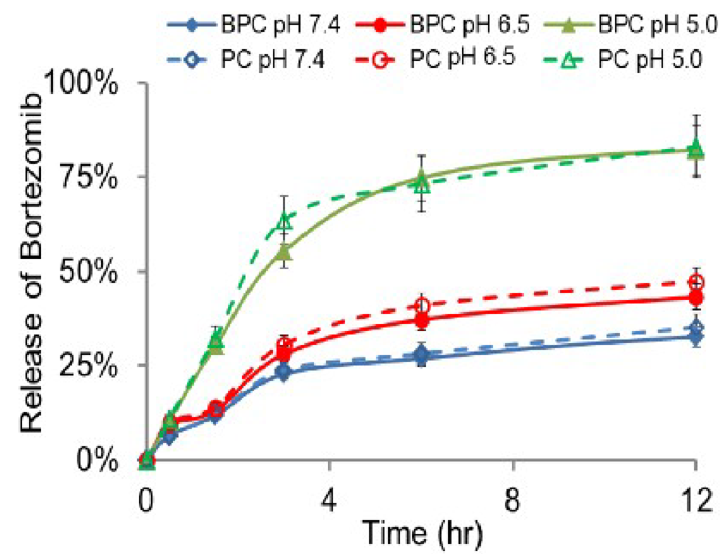


Bortezomib (BTZ)

Our catechol polymer design:







http://faculty.mccormick.northwestern.edu/phillip-messersmith/research/Research-F.html

Other pH sensitive moieties we learned??

Drug delivery pH sensitive

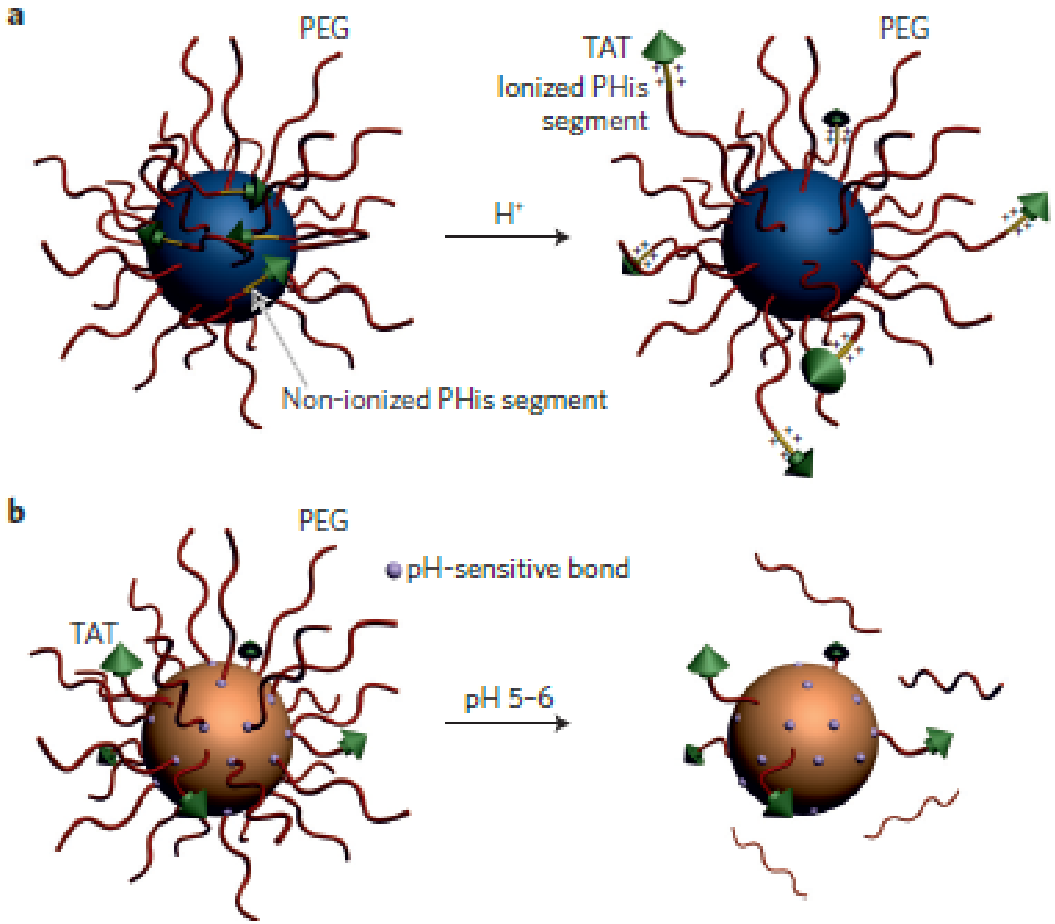
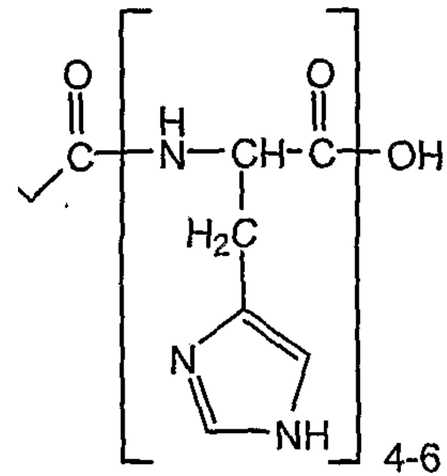


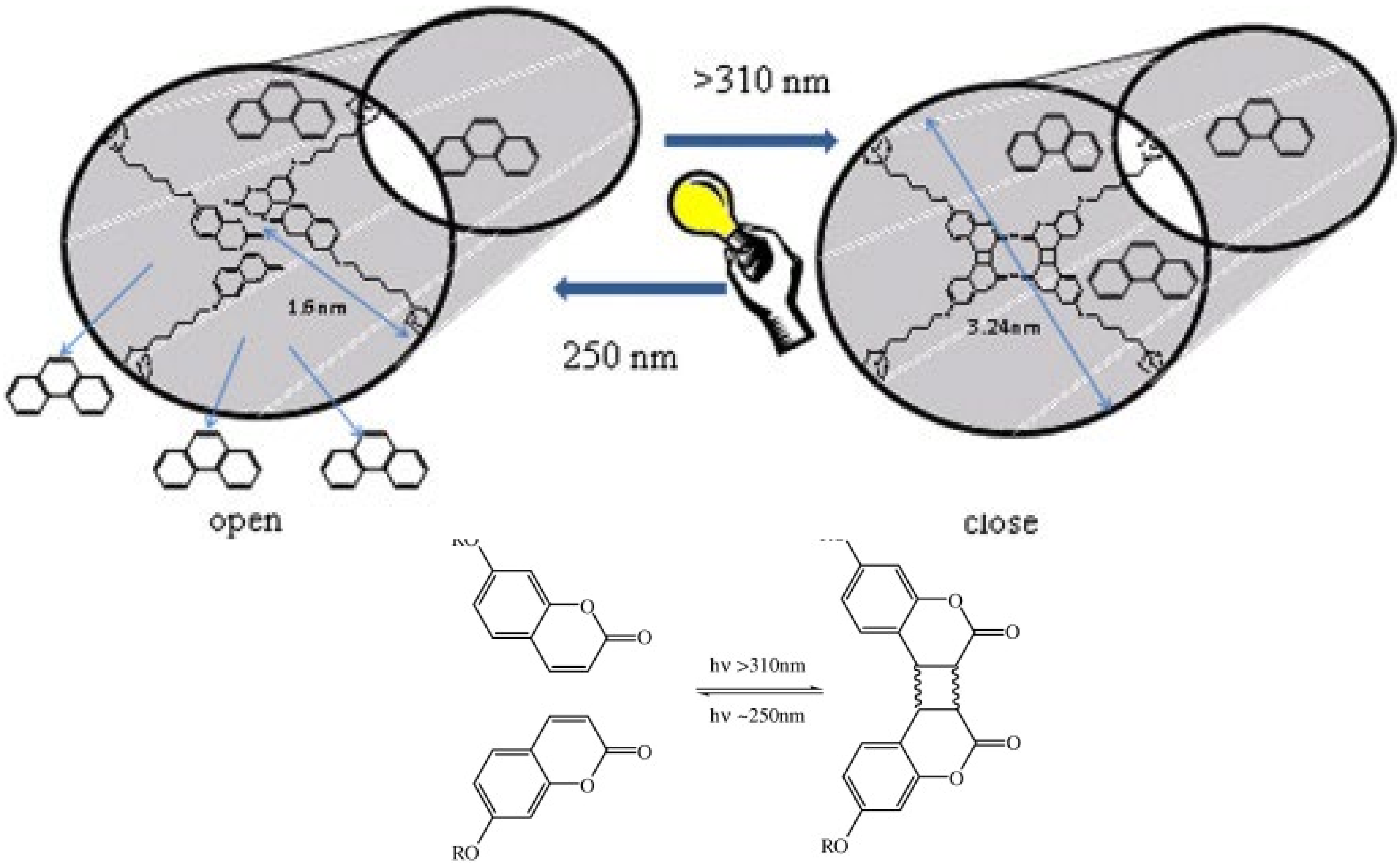
Figure 6 | pH-sensitive nanocarriers for efficient TAT-peptide exposure.
a, Polyhistidine (PHis)-based micelles responding to acidic tumour microenvironments by an efficient TAT-sequence exposure following ionization of the polyhistidine segments. **b**, TAT-peptide-decorated liposomes comprising a hydrolyzable PEG shell allowing improved exposure of the TAT sequence at low pH. Figure adapted with permission from: **a**, ref. 83, © 2008 Elsevier; **b**, ref. 84, © 2012 Elsevier.



Polyhistidine

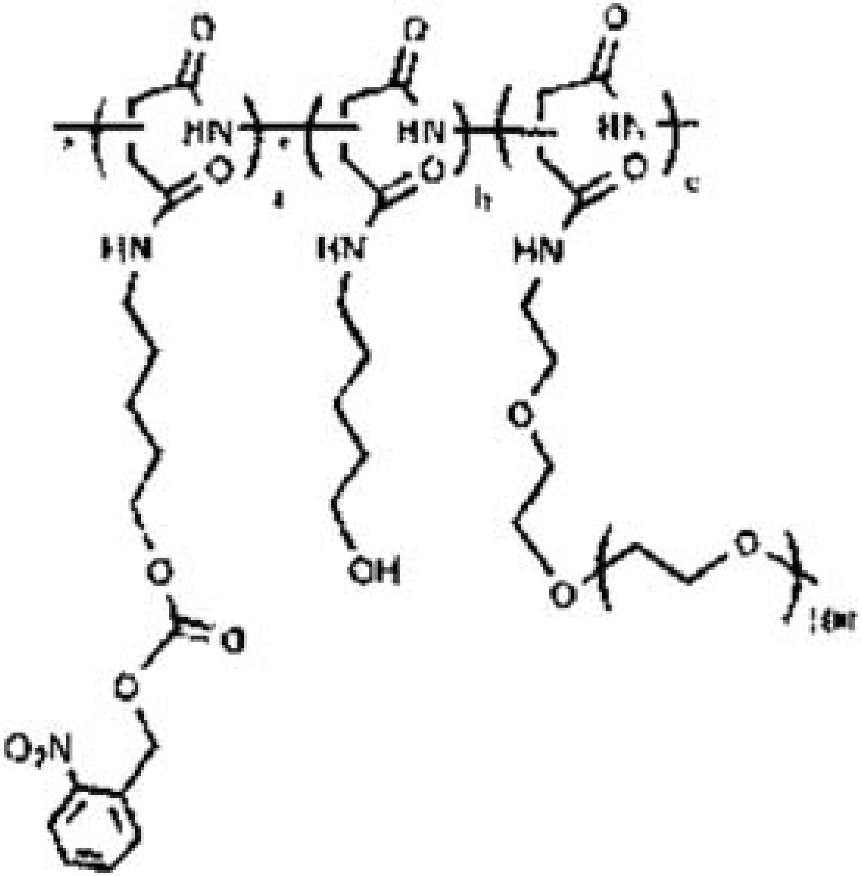
TAT peptide (YGRKKRRQRRR+C)
细胞穿膜肽

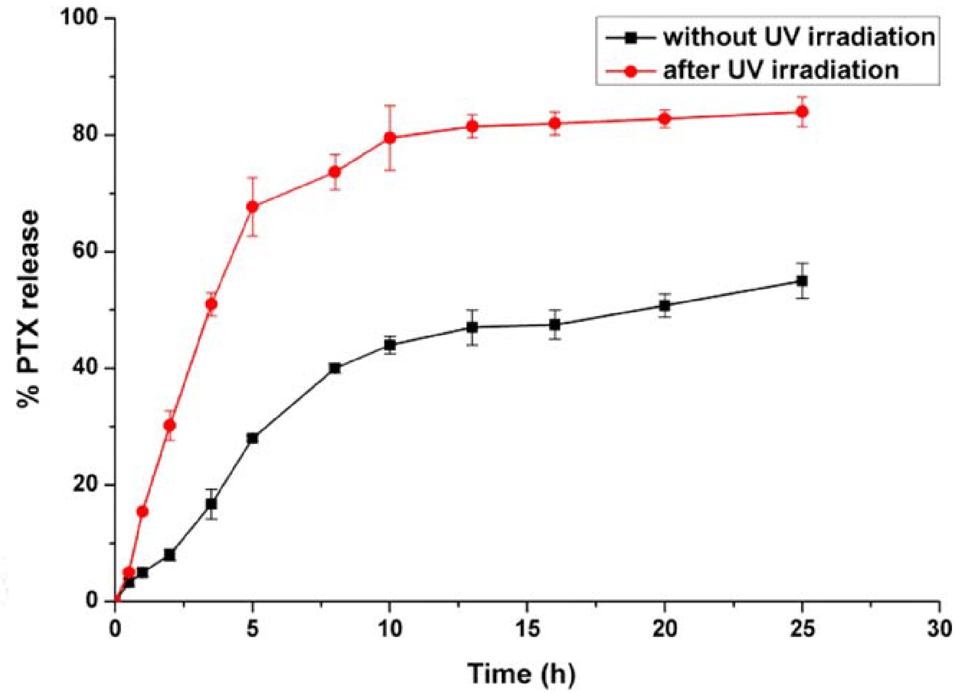
Drug delivery photo or light sensitive



https://www.sciencedirect.com/science/article/pii/S1742706110000759

Drug delivery photo or light sensitive





JOURNAL OF POLYMER SCIENCE, PART A: POLYMER CHEMISTRY 2016, 54, 2855–2863

Drug delivery

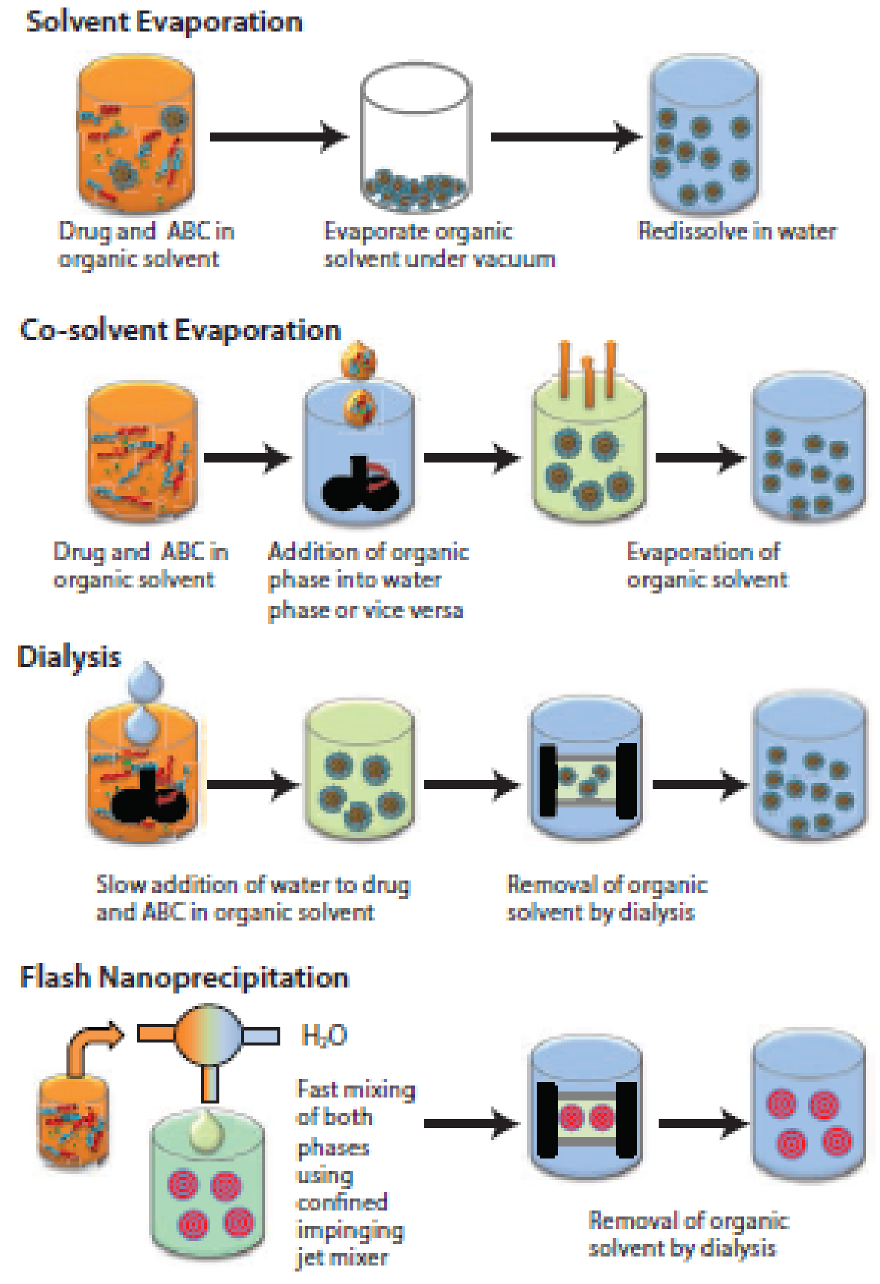


Figure 1. Techniques for drug loading of micelles

Many different approaches

Targeted delivery

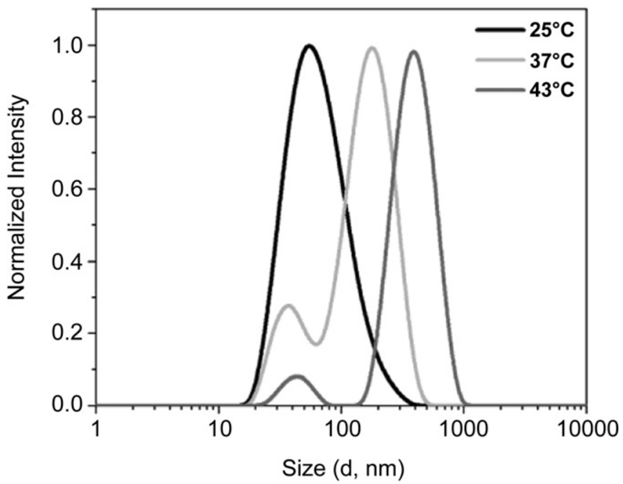
Image guided drug delivery

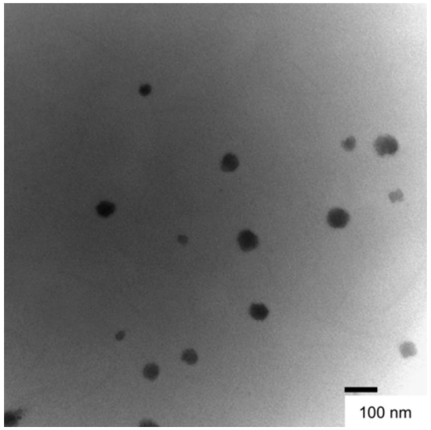
Drug delivery

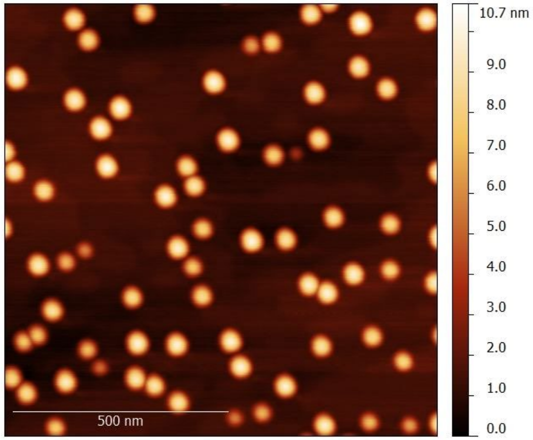
Structure characterization (^{1}H NMR, MS, FTIR, GPC, DSC)

Nanostructure Characterization

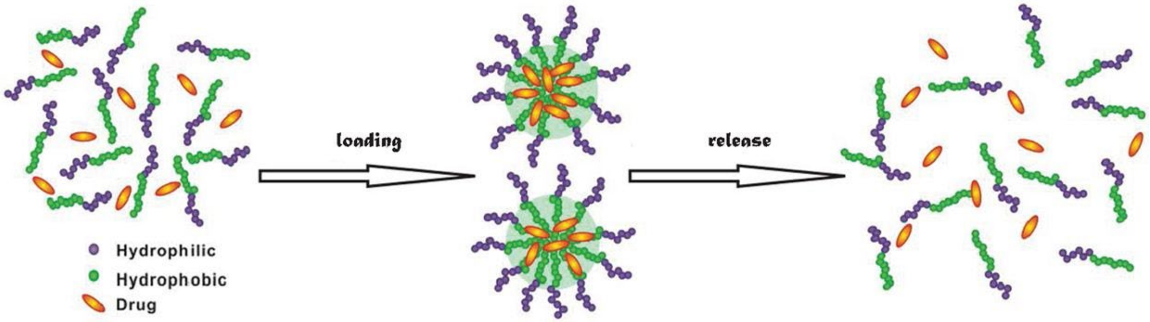
AFM, SEM, TEM; DLS;

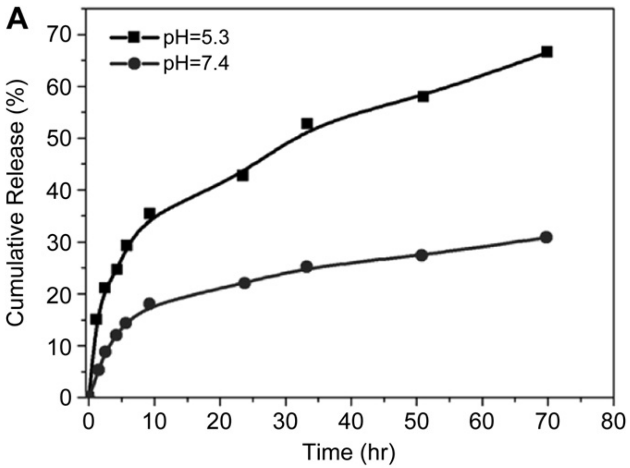






Drug loading and release



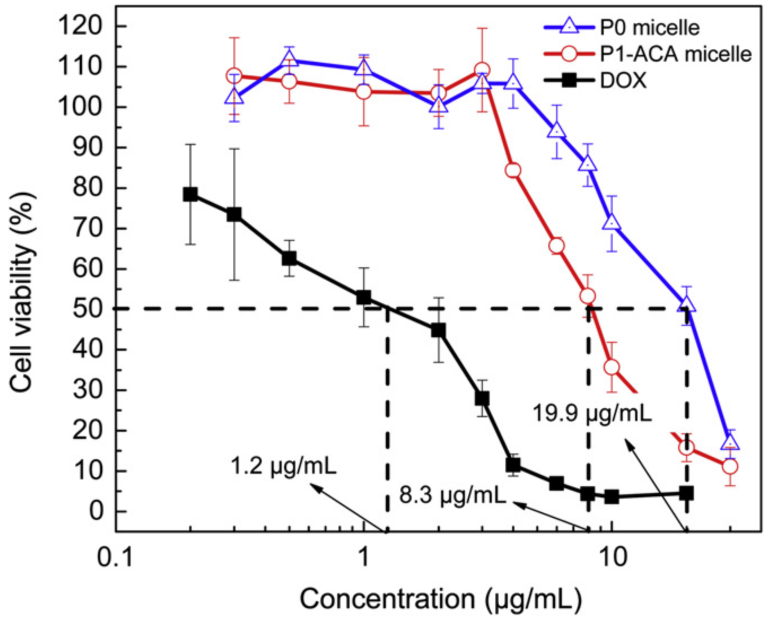


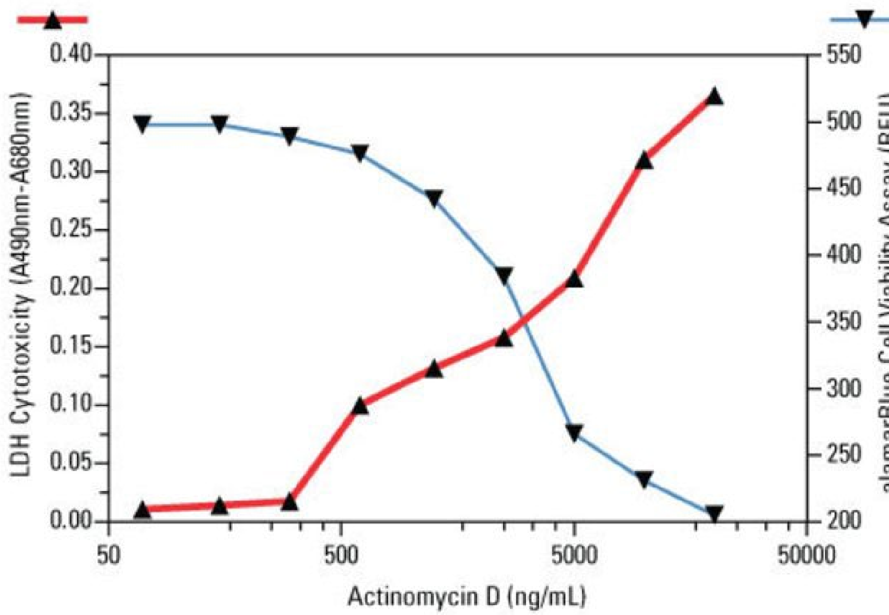
https://www.frontiersin.org/articles/10.3389/fmats.2015.00064/full

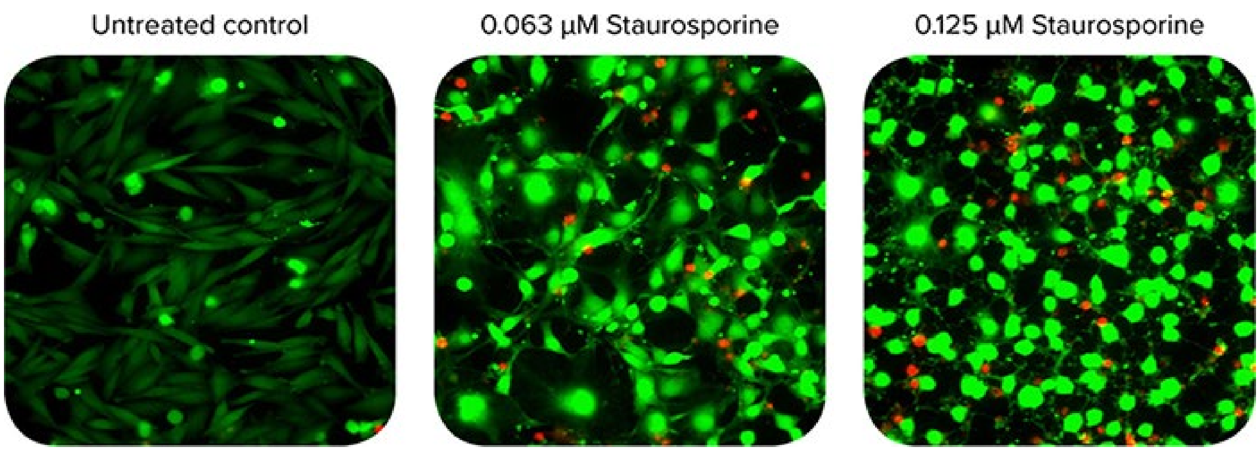
Drug delivery

Cytotoxicity evaluation

MTT, WST, LDH, Live-dead kits, flow-cytometry,







Drug delivery

In-vitro evaluation (drug distribution in cells)

Confocal microscope

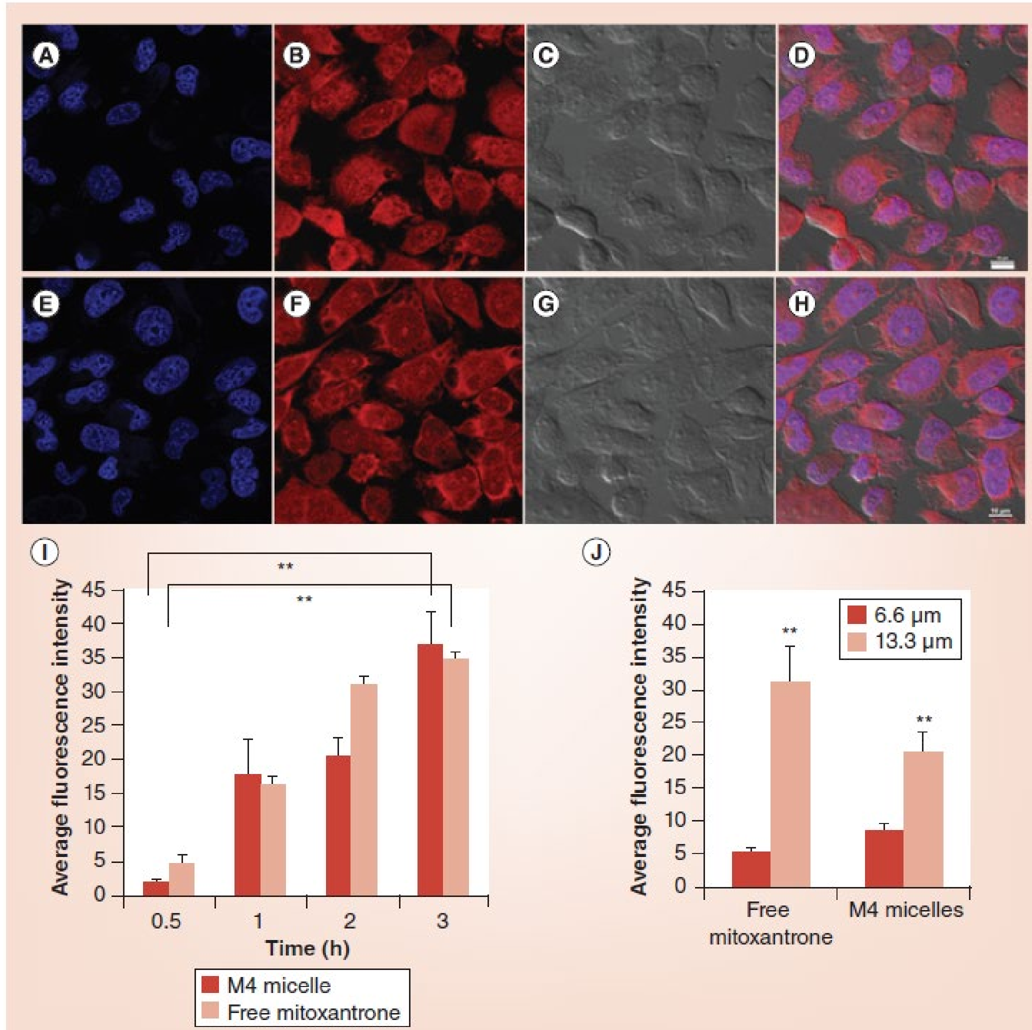
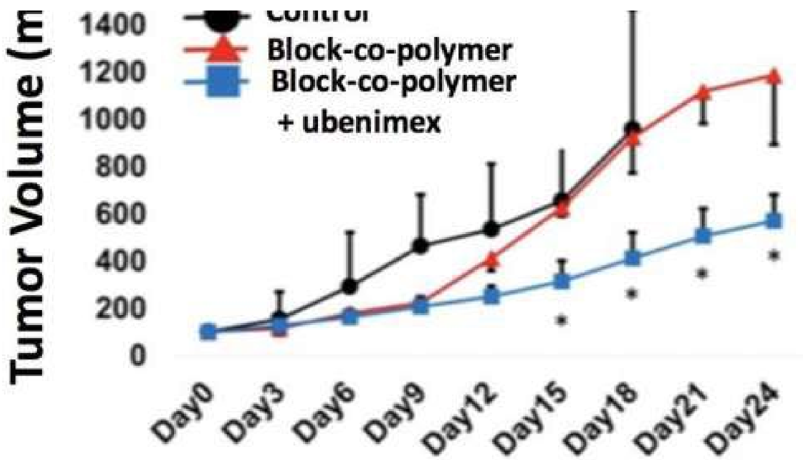


Figure 4. Confocal microscopy images showing internalization of 13.3 μM (A–D) M4 micelles and (E–H) free mitoxantrone in PC3 prostate cancer cells after 3 h. The nucleus was stained with DAPI **(A & E)** (blue; ex: 358 nm, em: 461 nm). Mitoxantrone **(B & F)** was excited with a

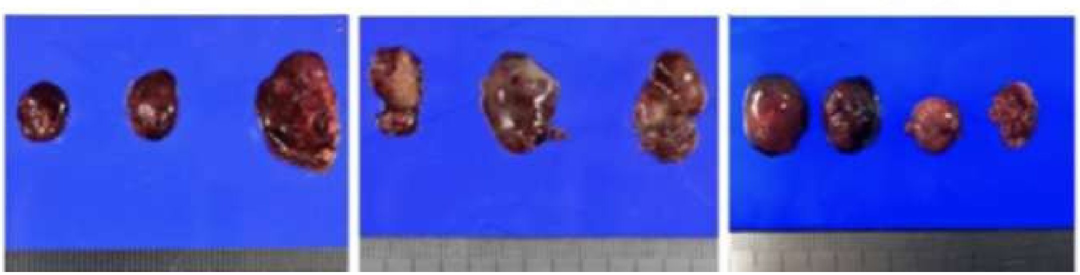
Drug delivery

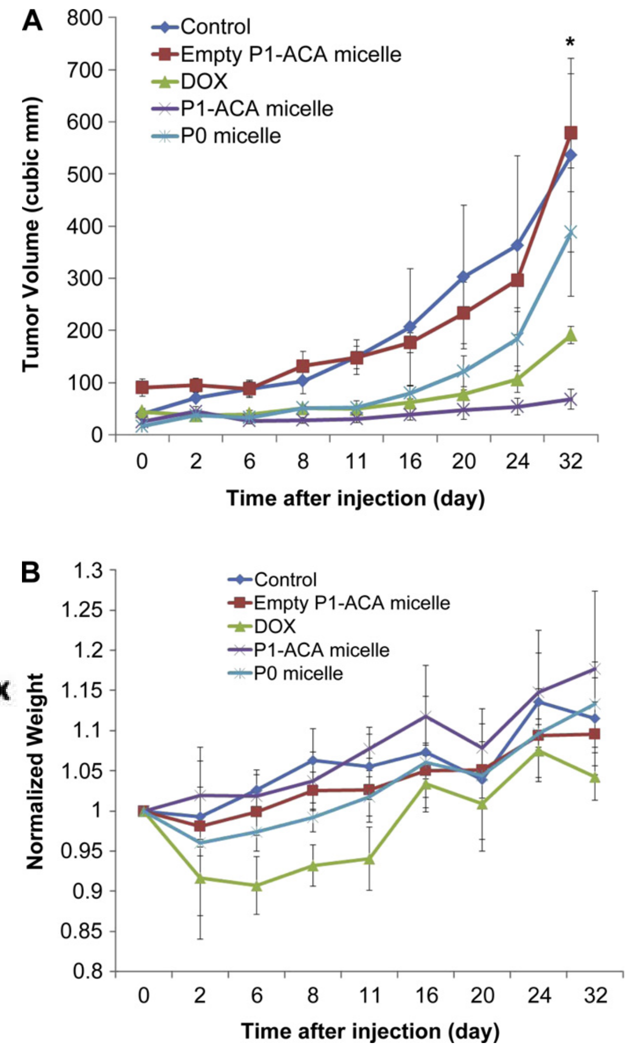
In-vivo evaluation

Animal images, tumor sizes, weight loss



Control Block-co-polymer Block-co-polymer + ubenimex





按药物作用机理分为五类：①细胞毒类药物：烷化剂类，由其氮芥基因作用于DNA和RNA、酶、蛋白质，导致细胞死亡。如氮芥、卡莫司汀（卡氮芥）、环磷酰胺、白消安（马利兰）、洛莫司汀（环己亚硝脲）等。②抗代谢类药：此类药物对核酸代谢物与酶结合反应有相互竞争作用，影响与阻断了核酸的合成。如氟尿嘧啶、甲氨喋呤、阿糖胞苷、巯基嘌呤、替加氟（呋喃氟尿嘧啶）等。③抗生素类：有抗肿瘤作用的如放线菌素D（更生霉素）、丝裂霉素、博菜霉素、阿毒素、平阳霉素、柔红霉素、光辉霉素等。④生物碱类：主要为干扰细胞内纺锤体的形成，使细胞停留在朋丝分裂中期。如长春新碱、长春碱、羟基树碱及鬼臼毒素类依托泊苷（VP-16）、替尼泊苷（VM-26）。⑤激素类，能改变内环境进而影响肿瘤生长，有的能增强机体对肿瘤侵害的抵抗力。常用的有他莫昔芬（三苯氧胺）、乙烯雌酚、黄体酮、丙酸睾丸酮、甲状腺素、泼尼松及地塞米松等。⑥其它：不属于以上诸类如甲基苄肼、羟基脲、L-门冬酰胺酶、顺铂、卡铂、抗癌锑、三嗪咪唑胺等。脂质体包裹5-氟尿嘧啶为导向性剂型。

按药物对细胞增殖周期的影响，可分为：①周期非特异性药物，即对增殖或非增殖细胞都有作用的药物，如氮芥类、环磷酰胺、抗生素类等。它们对小鼠骨髓干细胞和淋巴肿瘤细胞的量效曲线都呈指数性，其中氮芥和丝裂霉素选择性低（杀伤两类细胞的曲线斜率很接近），而大多数其他烷化剂选择性较高（表现于对两类细胞的量效曲线的斜率相差较大，见图2A，B）。②周期特异性药物，作用于细胞增殖整个或大部分周期时相者。如氟尿嘧啶等抗代谢类药物。③周期时相特异药物：药物选择性作用于某一个时相，如阿糖胞苷、羟基脲抑制S期，长春新碱对M相的抑制作用。这类药物对骨髓及瘤 细胞的量效曲线也随剂量增大而下降，但达到一定剂量时即向水平方向转折，成为一个坪，即再增加剂量，不再有更多的细胞被杀死。